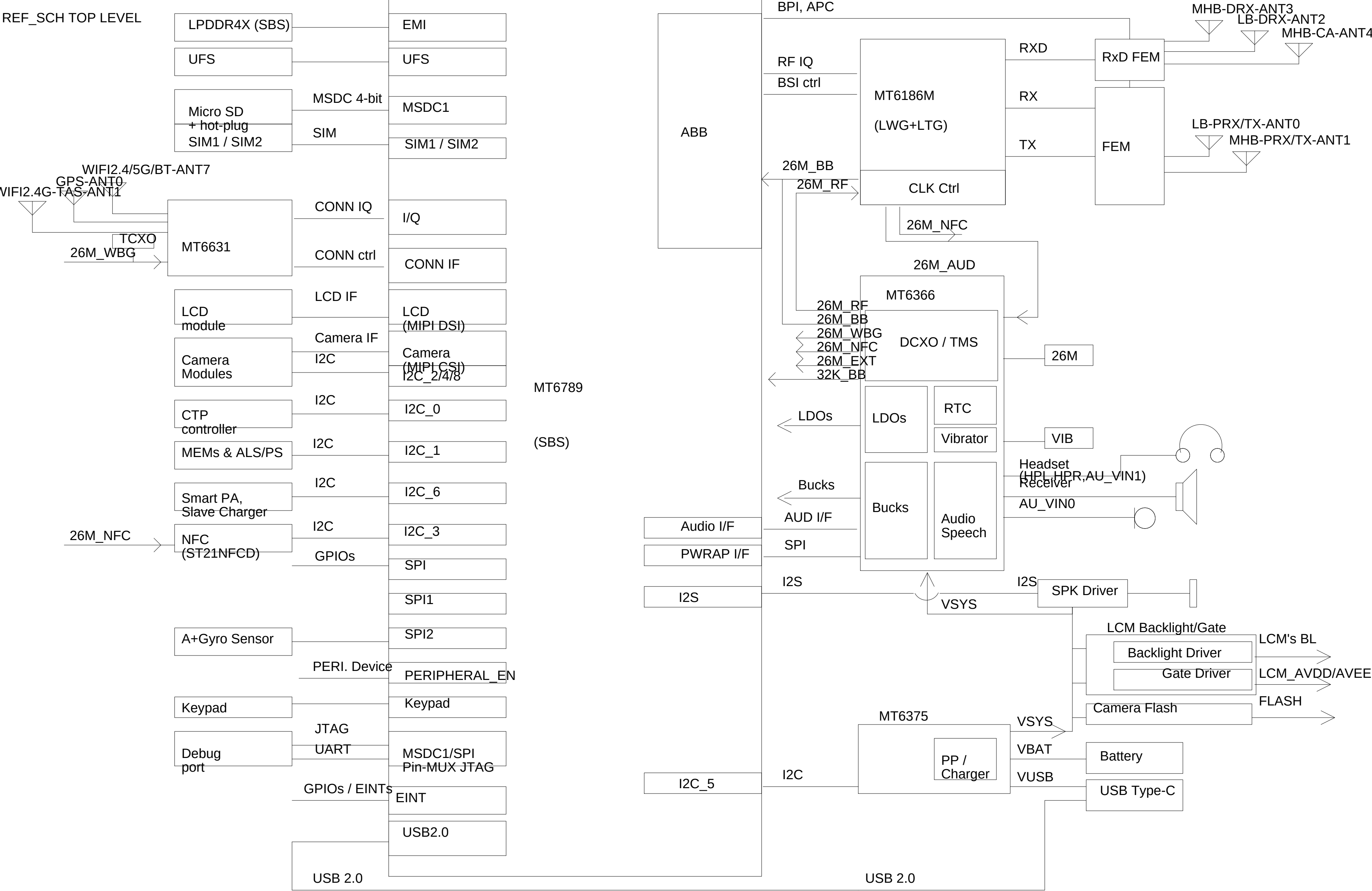


REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:

Project : MT6789, LPDDR4X (SBS)

REF_SCH TOP LEVEL

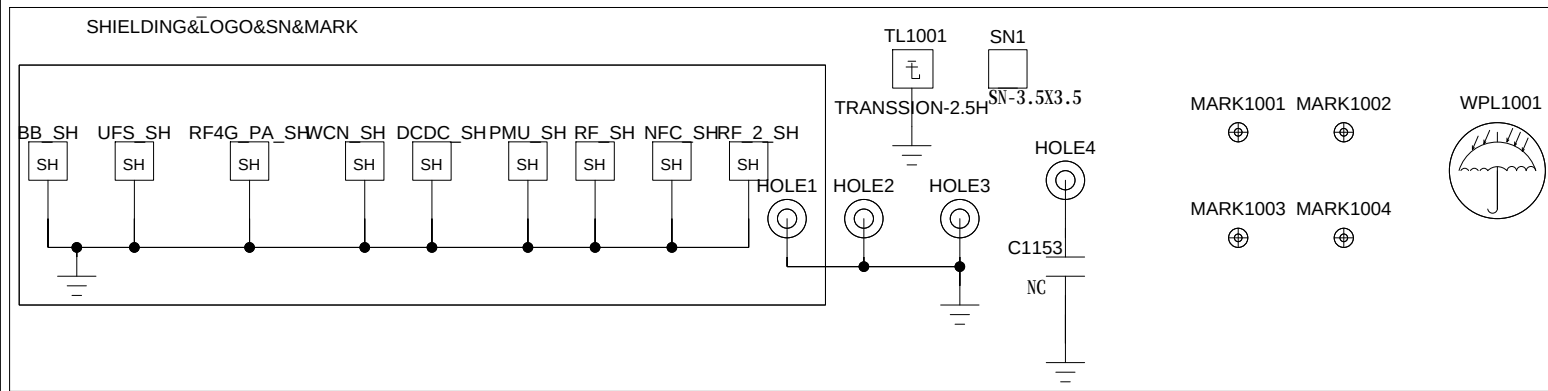
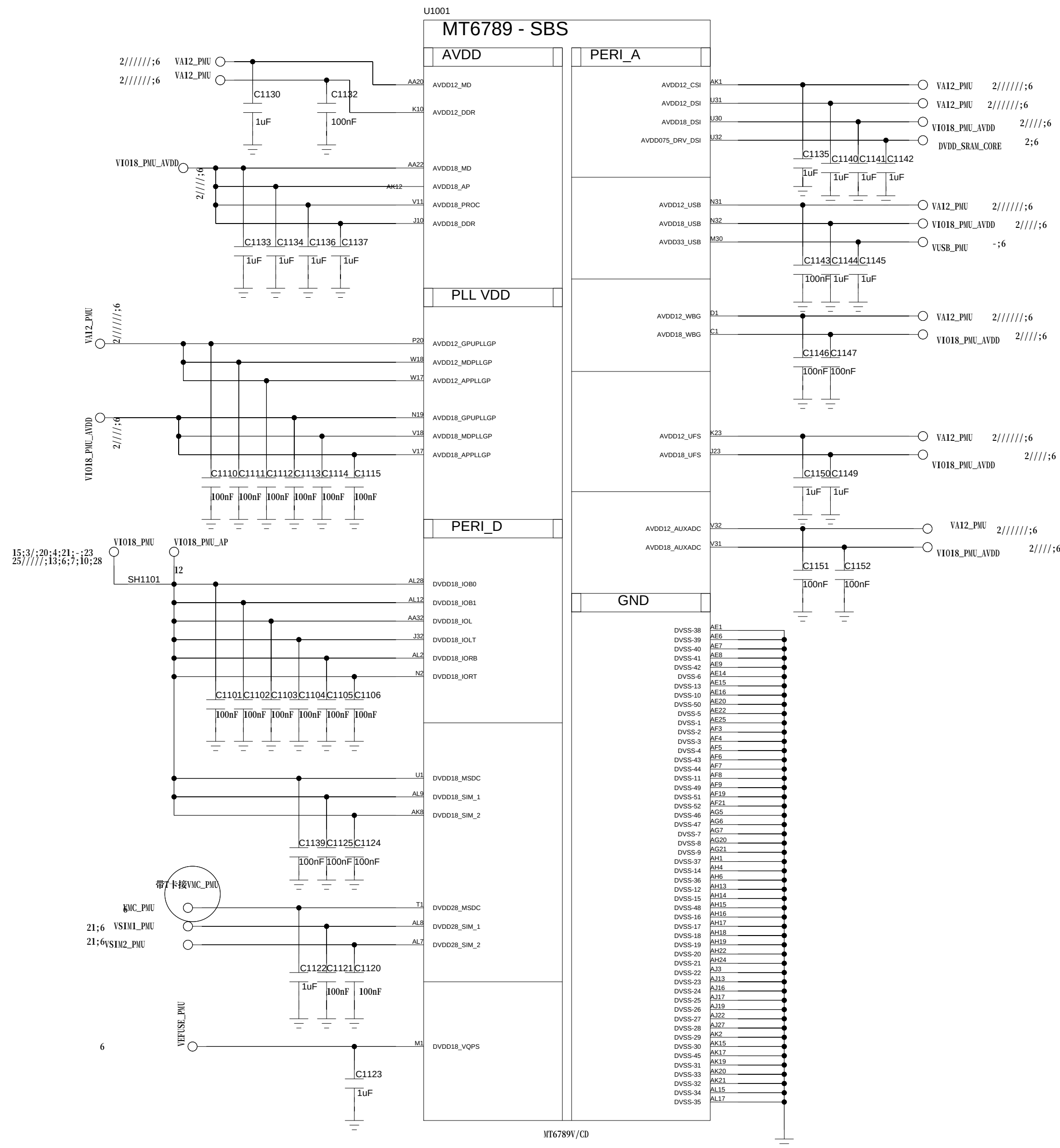
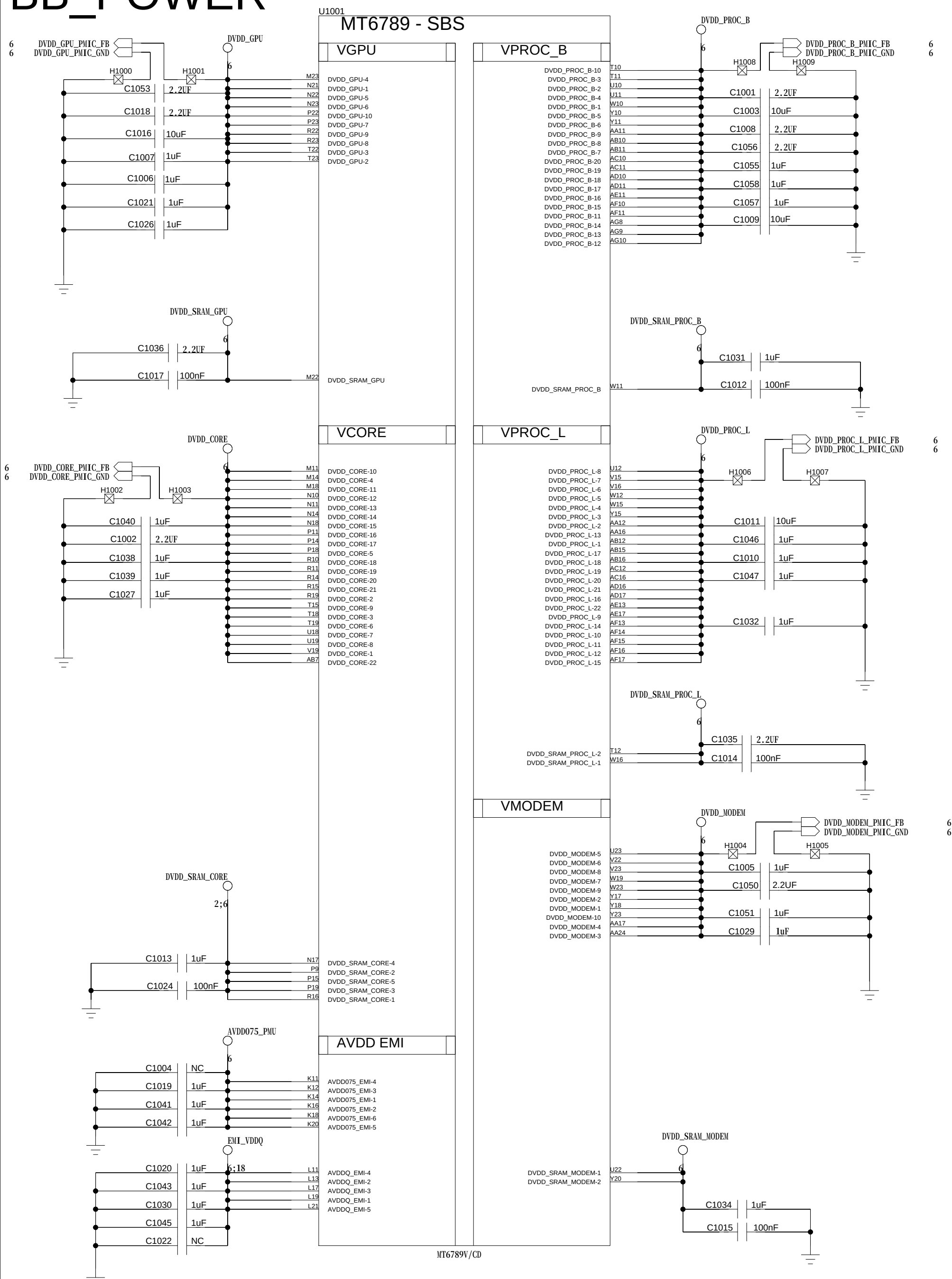
Pre-released for BOM Cost evaluation only



COMPANY: TRANSSION HOLDINGS				MODEL:G99_STD_MAIN		Modified Date: 03/03/2025:15:12	
DRAWN		DATED		TITLE: 00_BLOCK_DIAGRAM		VERSION:	SHEET: 1 OF 29
CHECKED		DATED	< >	Confidentiality	CONFIDENTIAL		

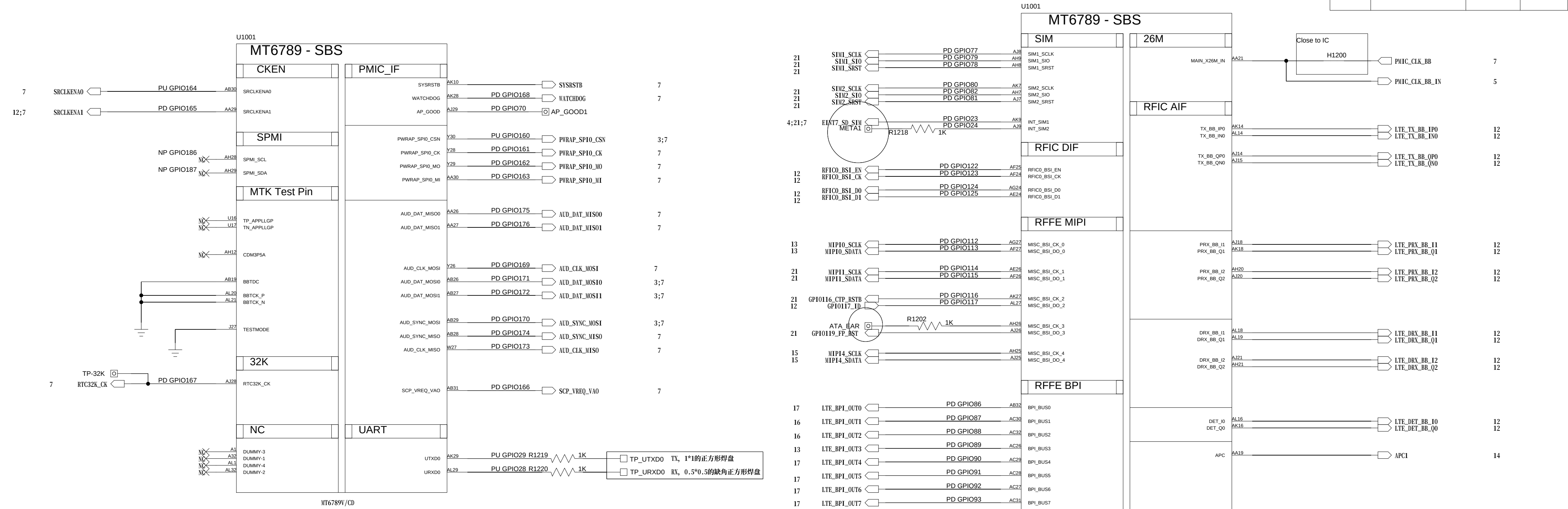
BB_POWER

REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:



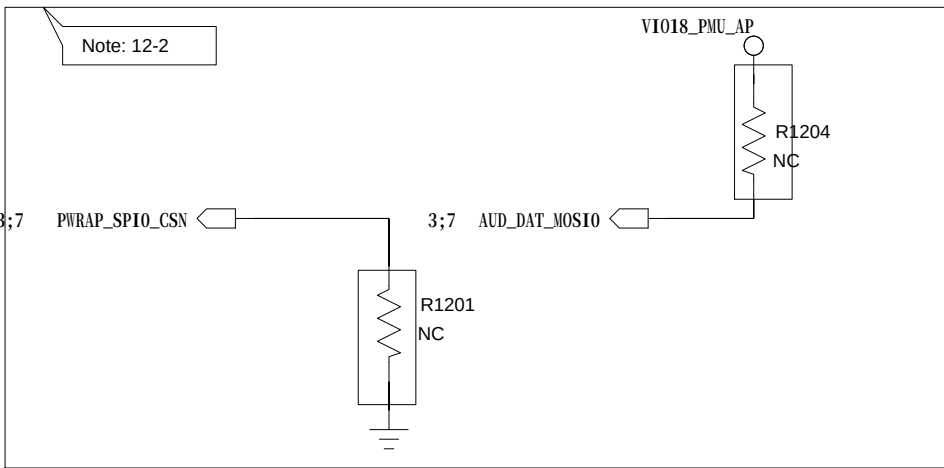
COMPANY: TRANSSION HOLDINGS				MODEL:G99_STD_MAIN		Modified Date: 03/03/2025:14:06	
DRAWN		DATED		TITLE: 10_BB_POWER_PDN		VERSION:	SHEET: 2 of 29
CHECKED		DATED	< >	Confidentiality	CONFIDENTIAL		

REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:

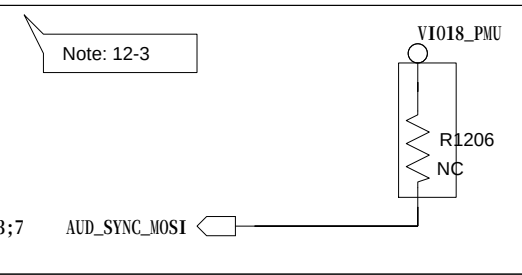


Schematic design notice of "12_BB_1" page:

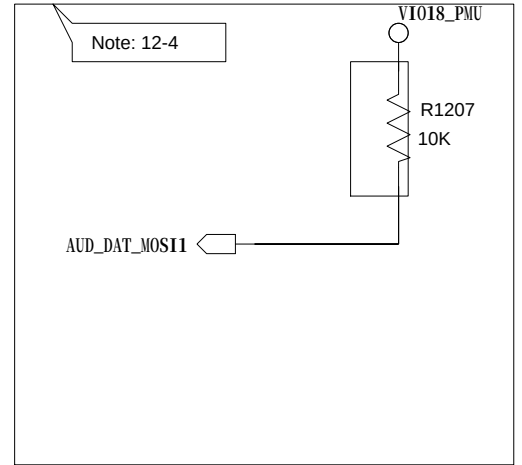
Note 12-2: "PWRAP_SPI0_CSN" and "AUD_DAT_MOSI0" pin features in trapping pin to enable JTAG.



PWRAP_SPI0_CSN AUD_DAT_MOSI0		AP JTAG Mode	IO JTAG Mode
H (Default)	L (Default)	N/A	N/A
H	H (by external PU)	SPI5_CSB, SPI5_CLK, SPI5_MO, SPI5_MI, EINT13	N/A
L (by external PD)	L	SPI5_CSB, SPI5_CLK, SPI5_MO, SPI5_MI, EINT13	SPI1_CSB, SPI1_CLK, SPI1_MO, SPI1_MI, EINT14
L (by external PD)	H (by external PU)	N/A	N/A

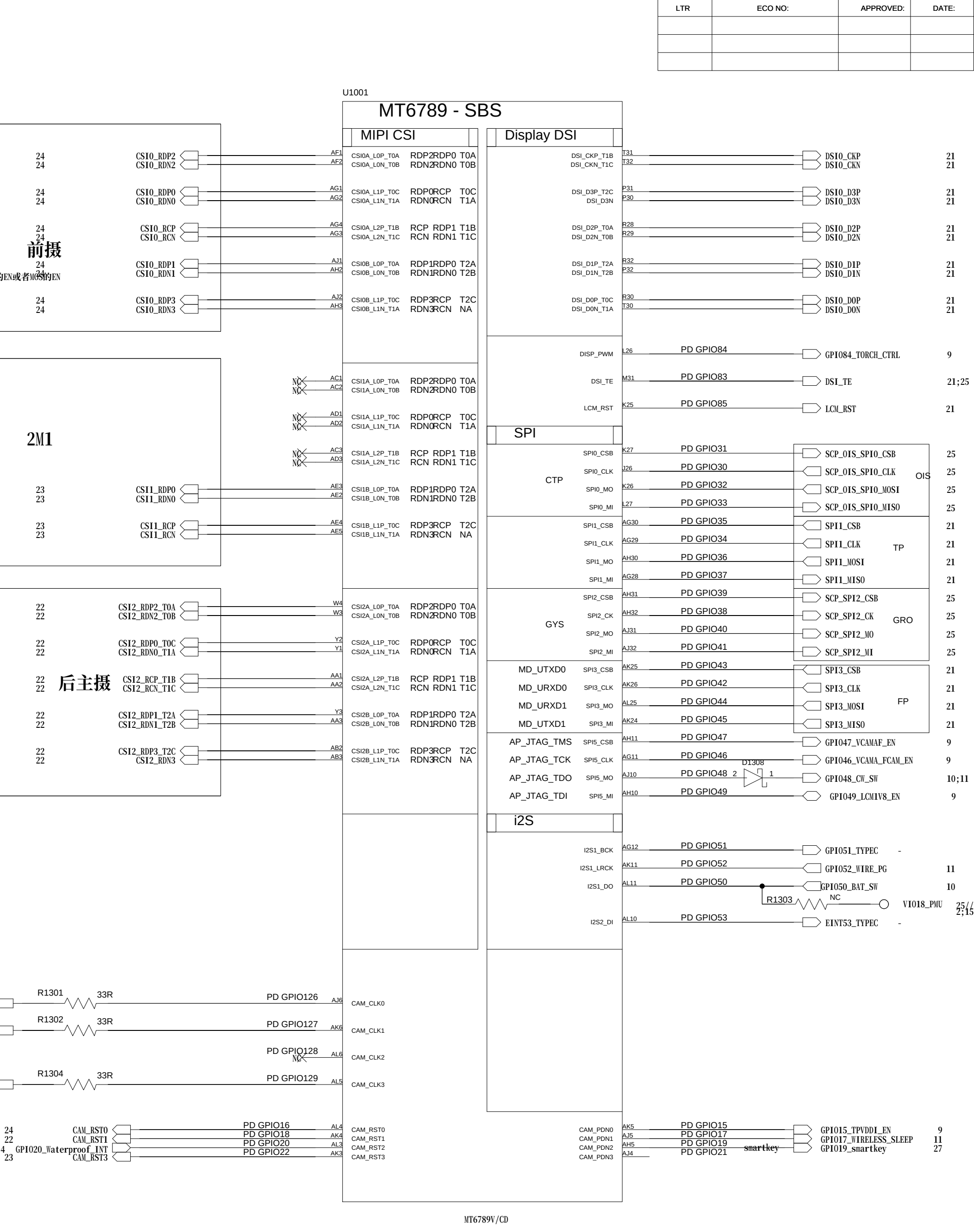
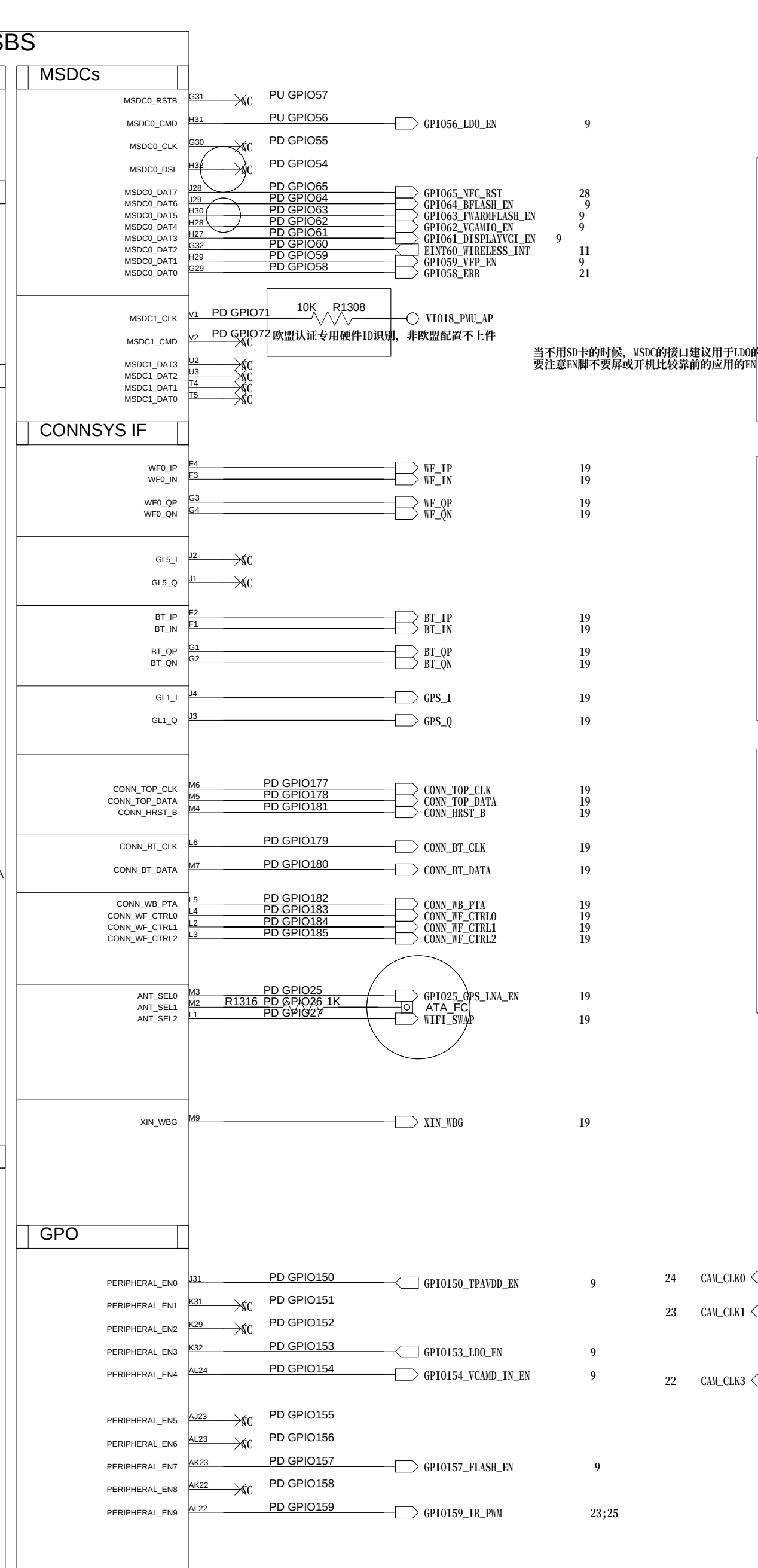
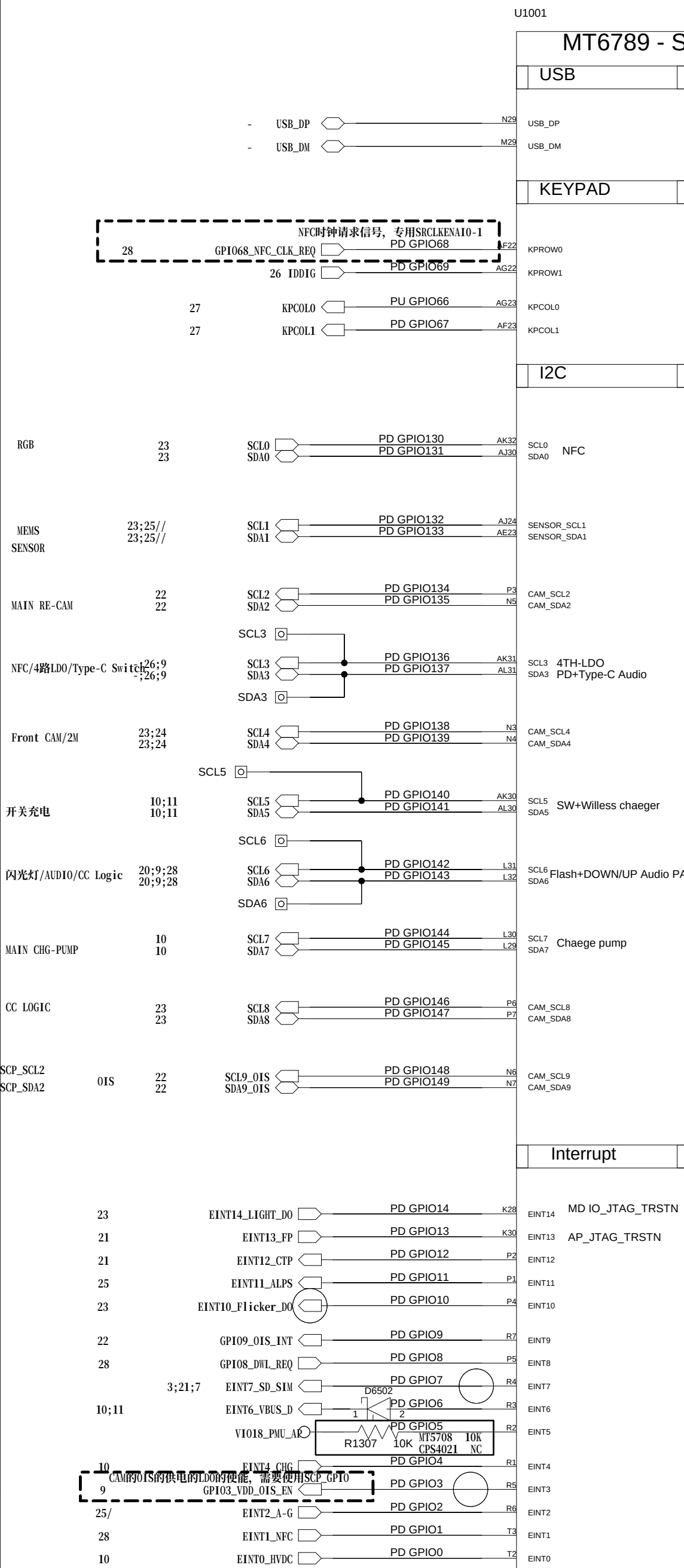


AUD_DAT_MOSI1		Mode
L (Default)		LP4x-MCP
H (by external PU)		LP4x-Discrete



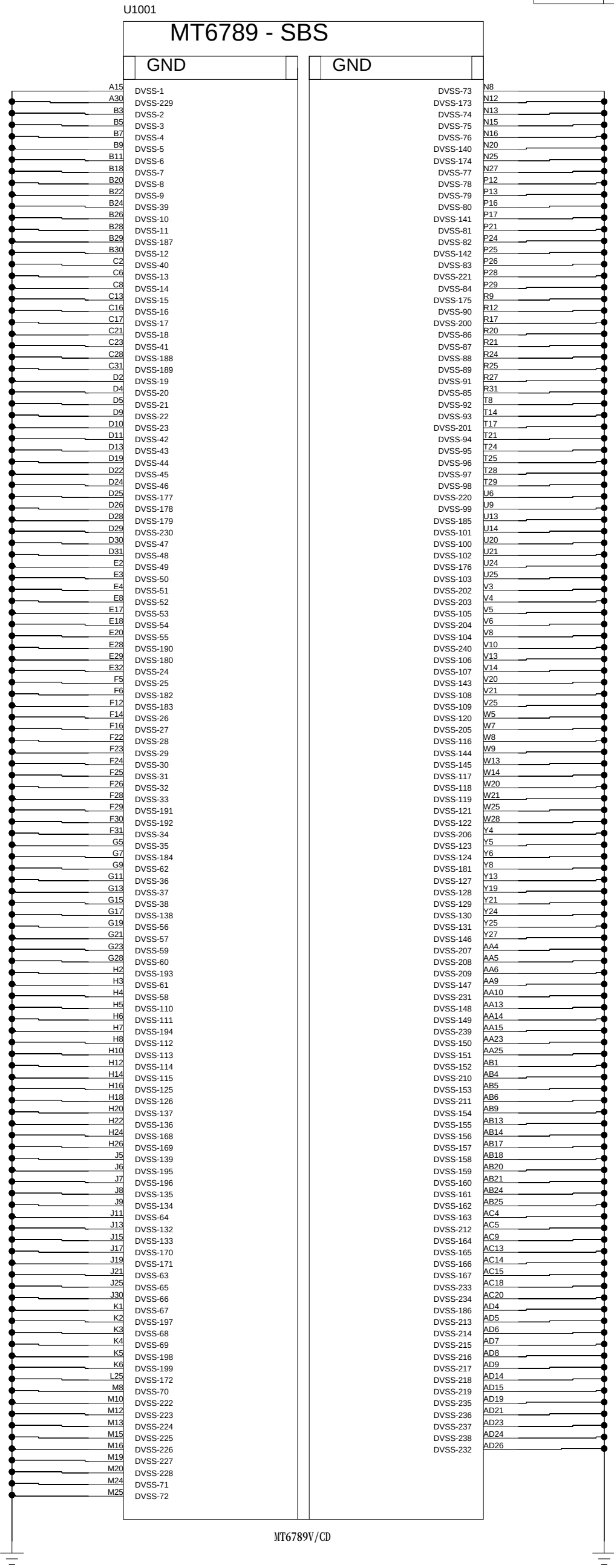
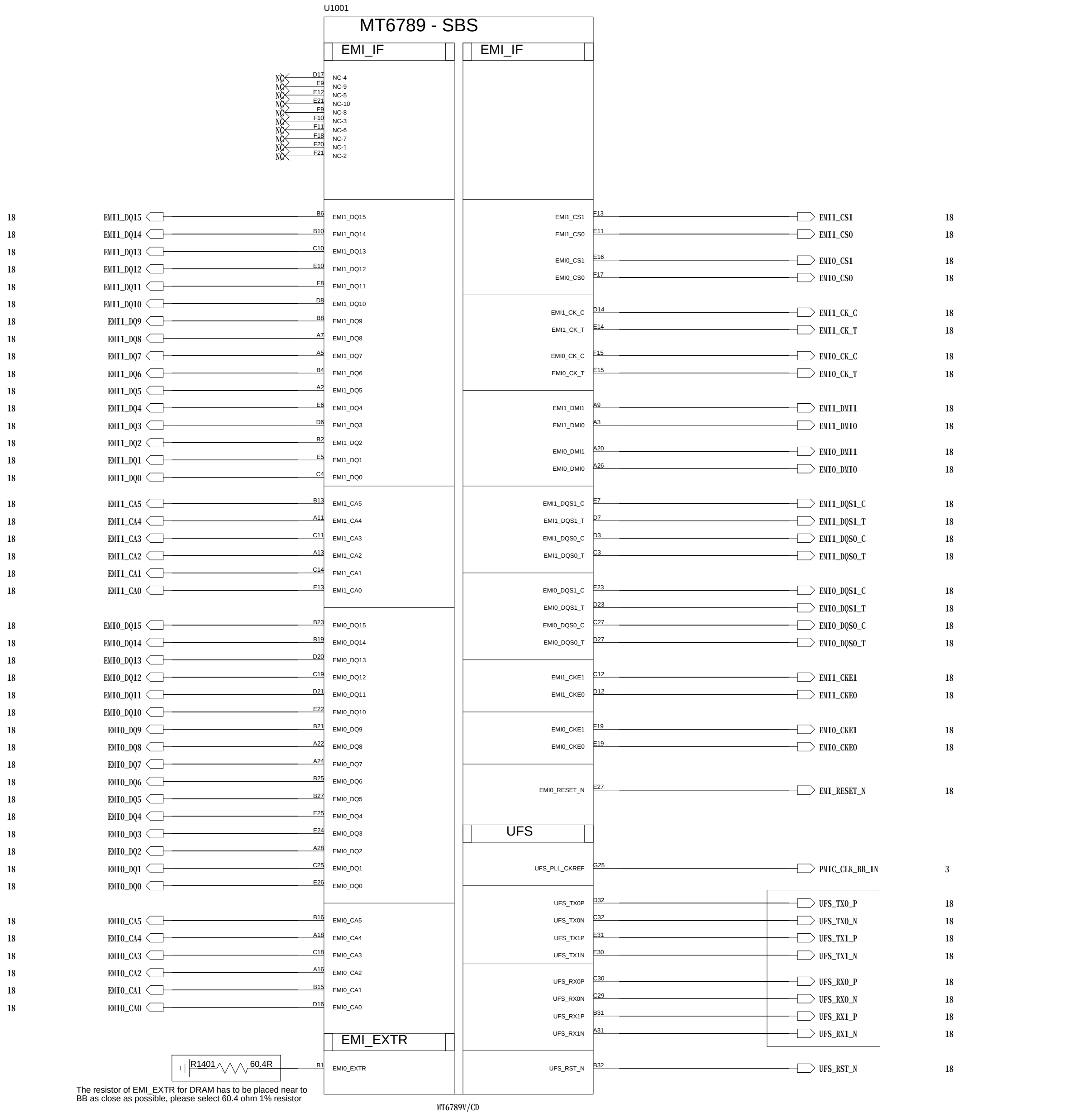
COMPANY: TRANSSION HOLDINGS				MODEL:G99_STD_MAIN		Modified Date: 01/04/2025:14:14	
DRAWN		DATED		TITLE: 12_BB_I_PMIC_RF		VERSION:	SHEET: 3 OF 29
CHECKED		DATED	< >	Confidentiality	CONFIDENTIAL		

REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:



COMPANY: TRANSSION HOLDINGS				MODEL:G99_STD_MAIN		Modified Date: 01/04/2025:14:11	
DRAWN		DATED		TITLE: 13_BB_II_CONN_RF		VERSION:	SHEET: 4 OF 29
CHECKED		DATED	< >	Confidentiality	CONFIDENTIAL		

REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:



COMPANY: TRANSSION HOLDINGS

MODEL:G99_STD_MAIN

Modified Date:

03/03/2025:14:06

DRAWN

DATED

TITLE: 14_BB_III_DDR

VERSION:

SHEET: 5 OF 29

CHECKED

DATED

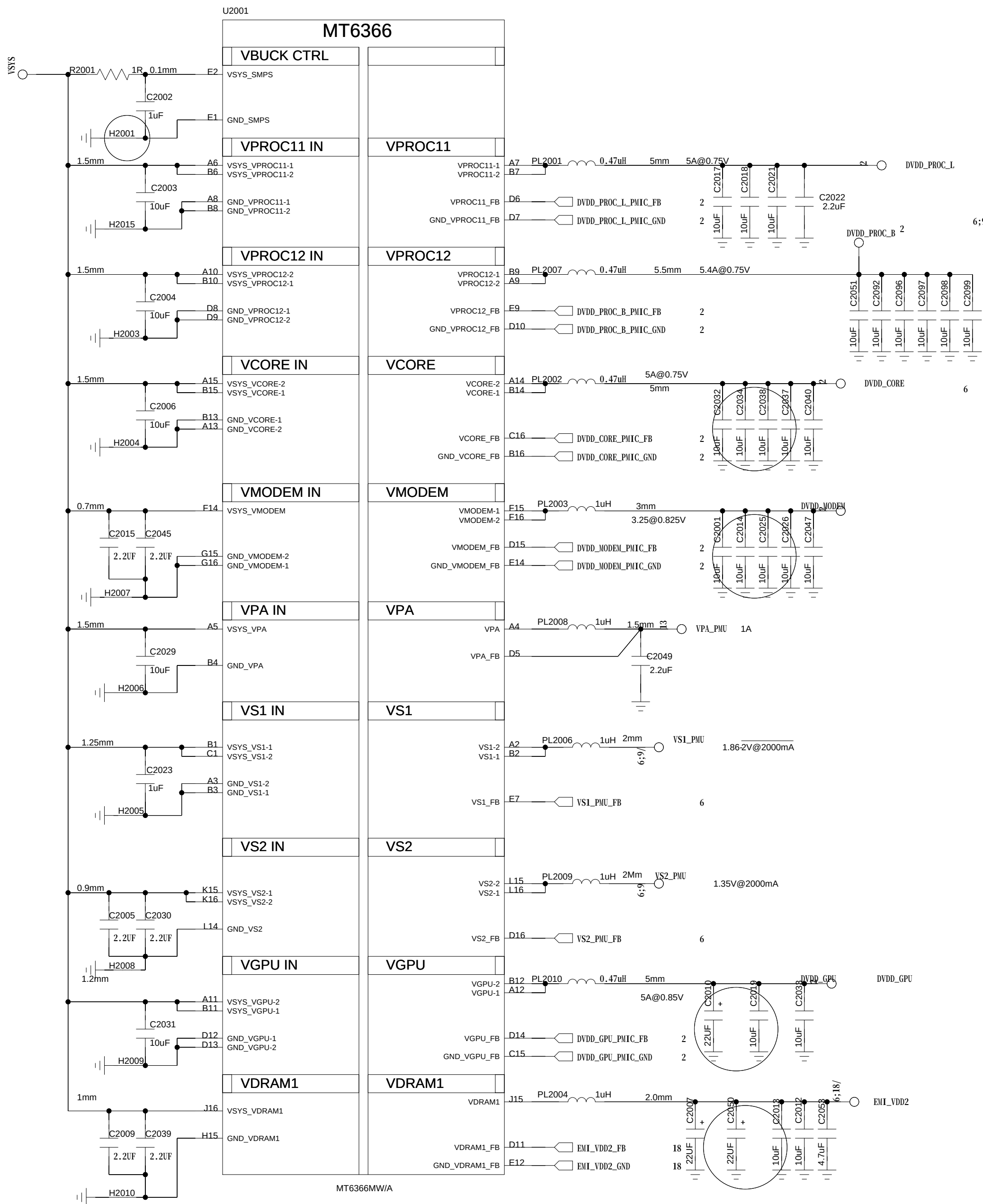
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Confidentiality

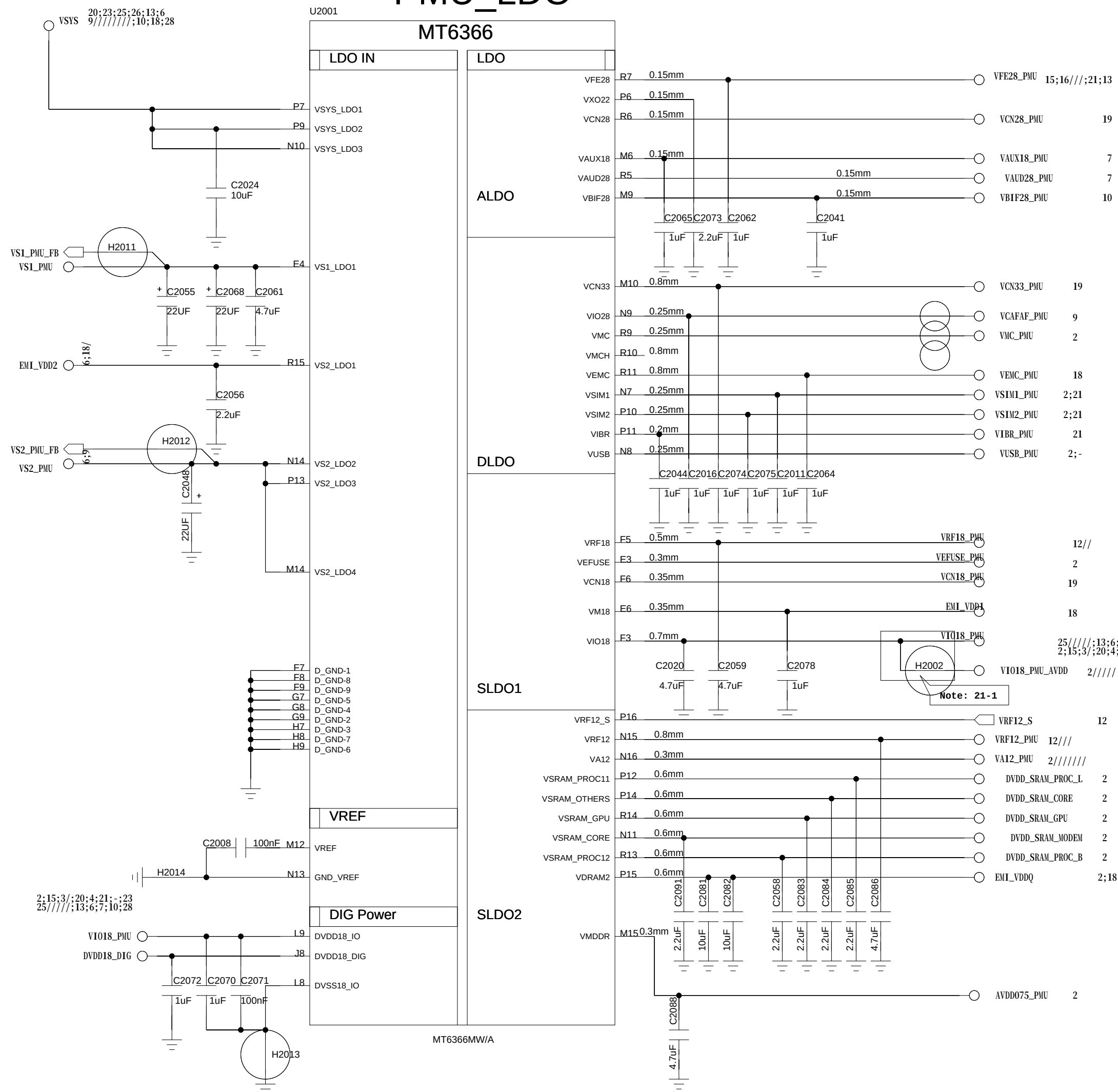
CONFIDENTIAL

POWER_MT6366_I

PMU_BUCK



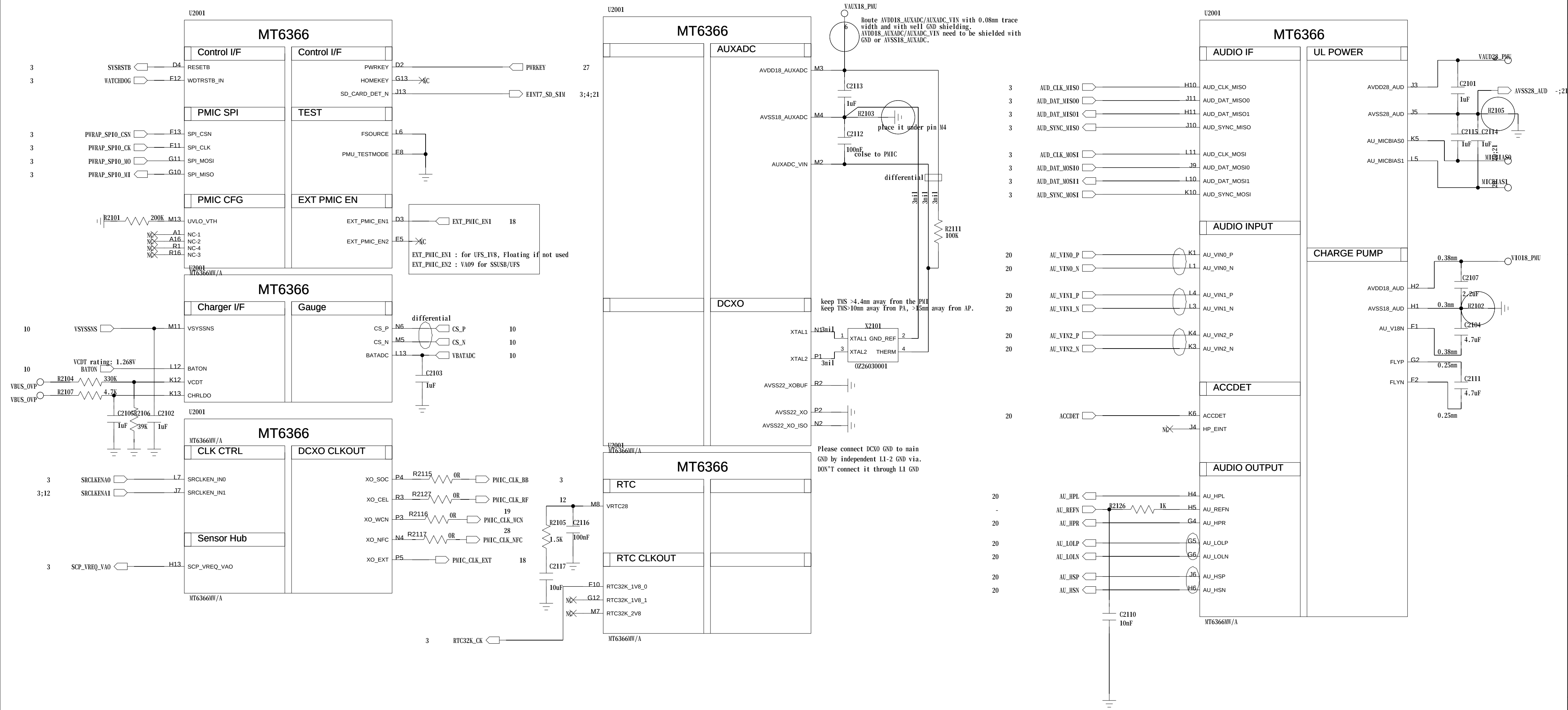
PMU_LDO



COMPANY: TRANSSION HOLDINGS				MODEL:G99_STD_MAIN		Modified Date: 04/03/2025:22:05	
DRAWN		DATED		TITLE: 20_POWER_MT6366_BUCK_LDO		VERSION:	SHEET: 6 OF 29
CHECKED		DATED	< >	Confidentiality	CONFIDENTIAL		

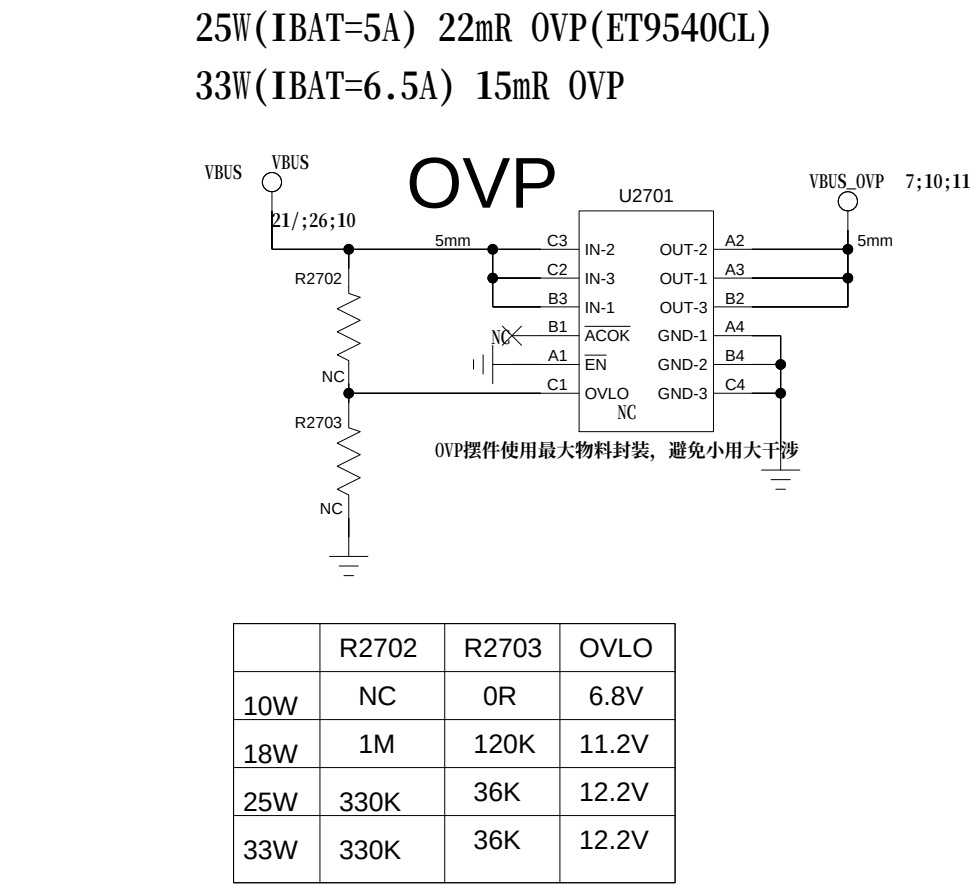
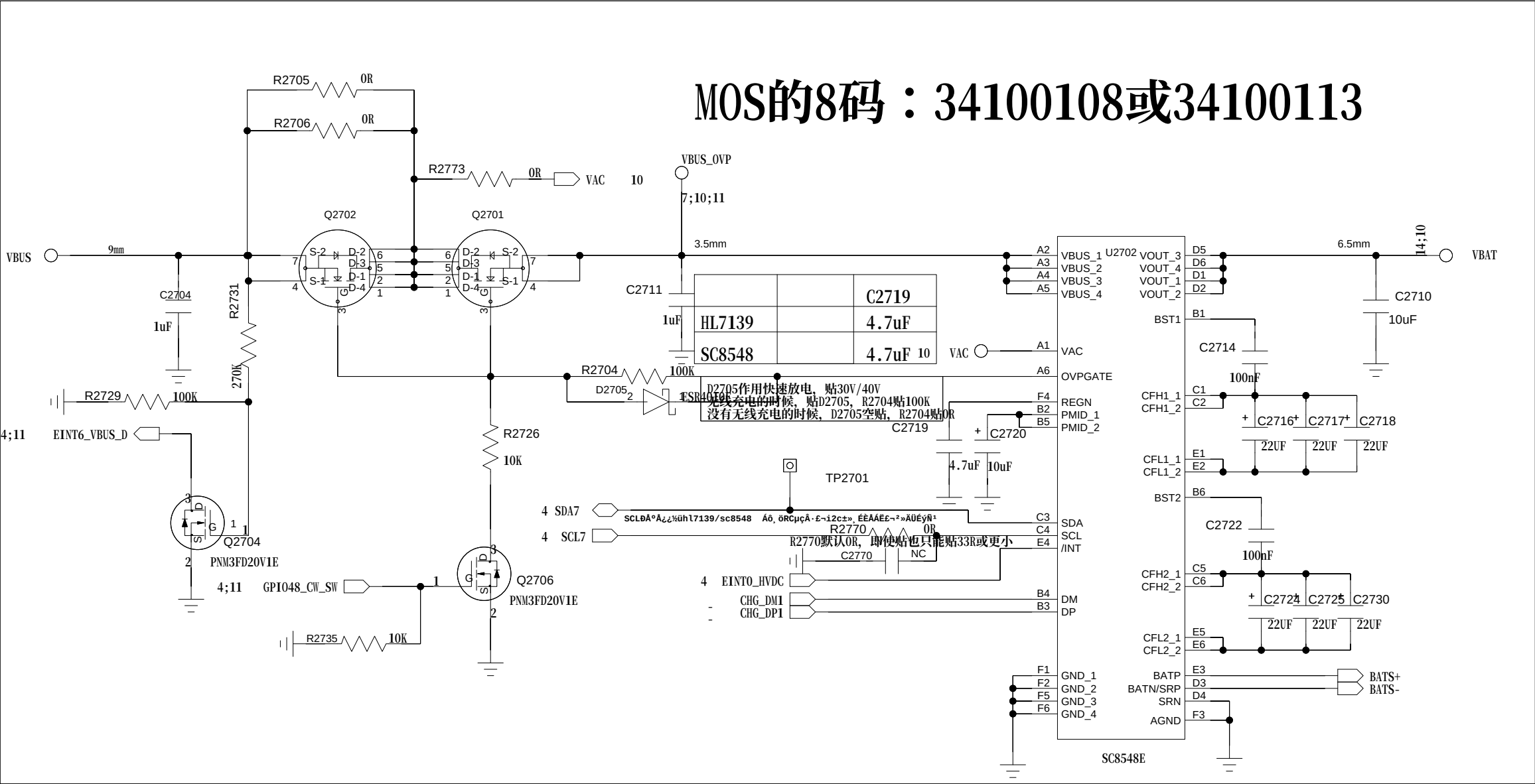
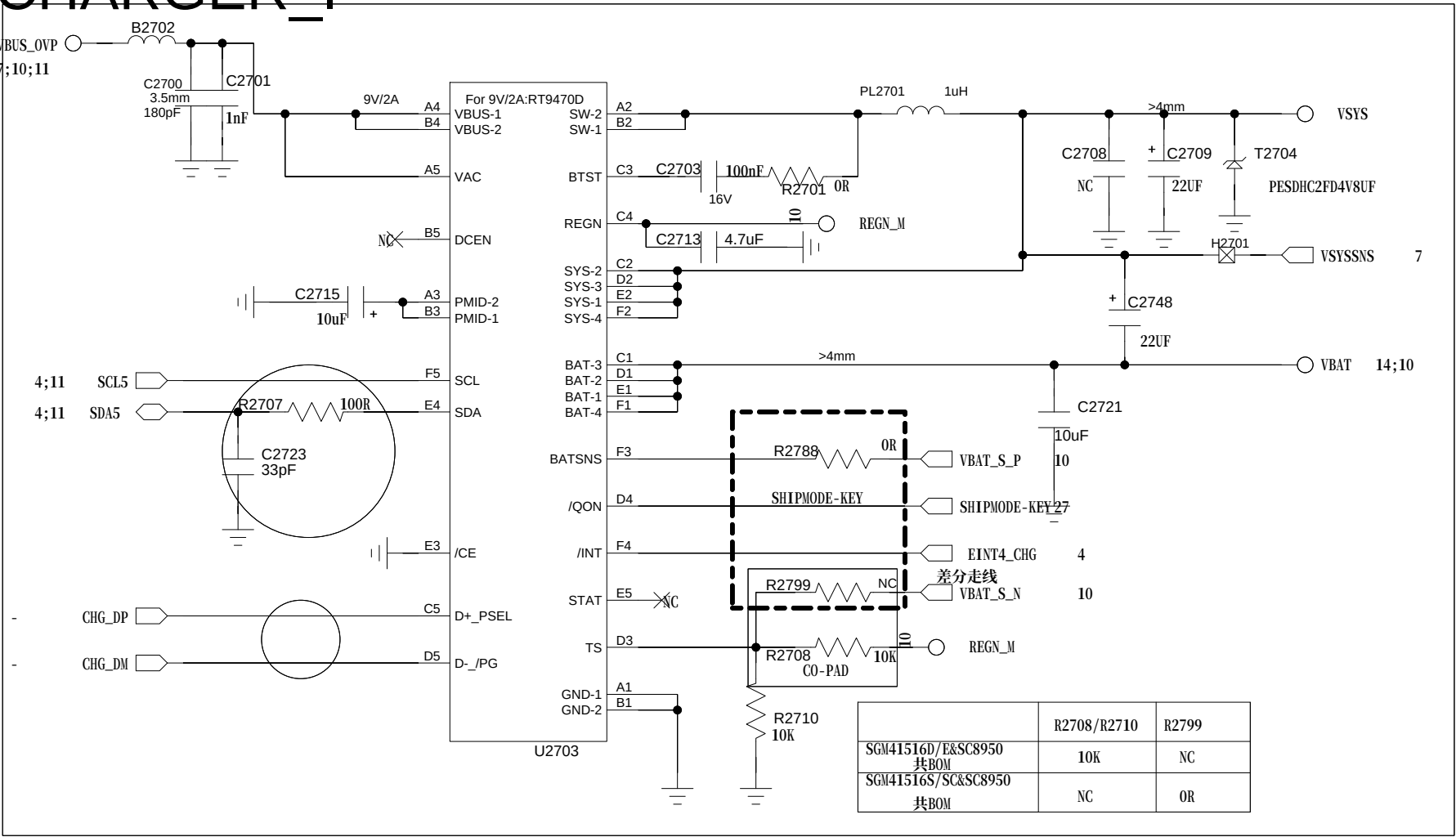
POWER_MT6366_II

REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:

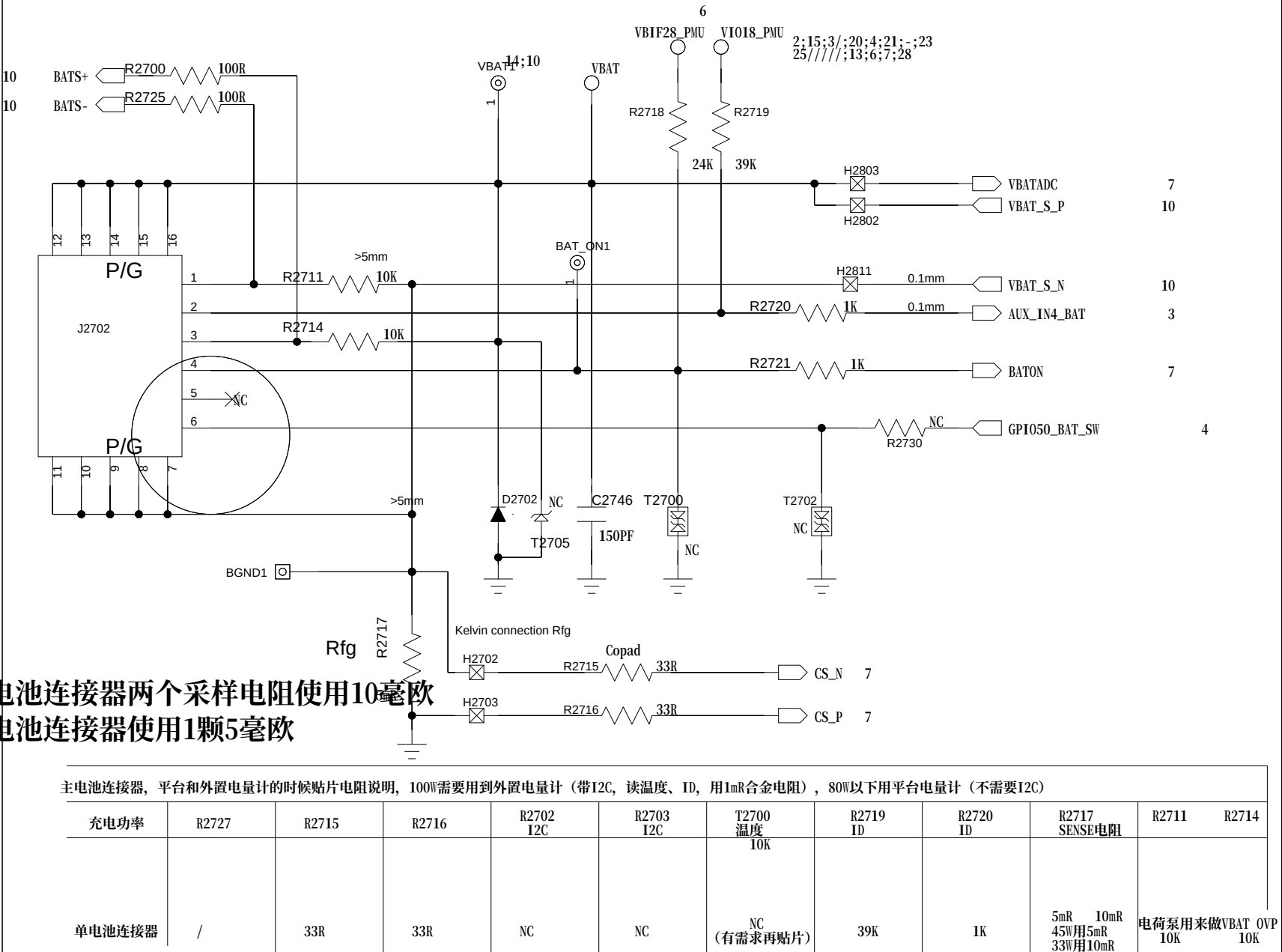


COMPANY: TRANSSION HOLDINGS				MODEL:G99_STD_MAIN		Modified Date: 03/03/2025:14:06	
DRAWN		DATED		TITLE: 21_POWER_MT6366_GENERAL_CLK		VERSION:	SHEET: 7 OF 29
CHECKED		DATED	< >	Confidentiality	CONFIDENTIAL		

POWER_THIRD-PARTY_II （此基线有线充电设计按照45W单电荷泵设计）
CHARGER_I



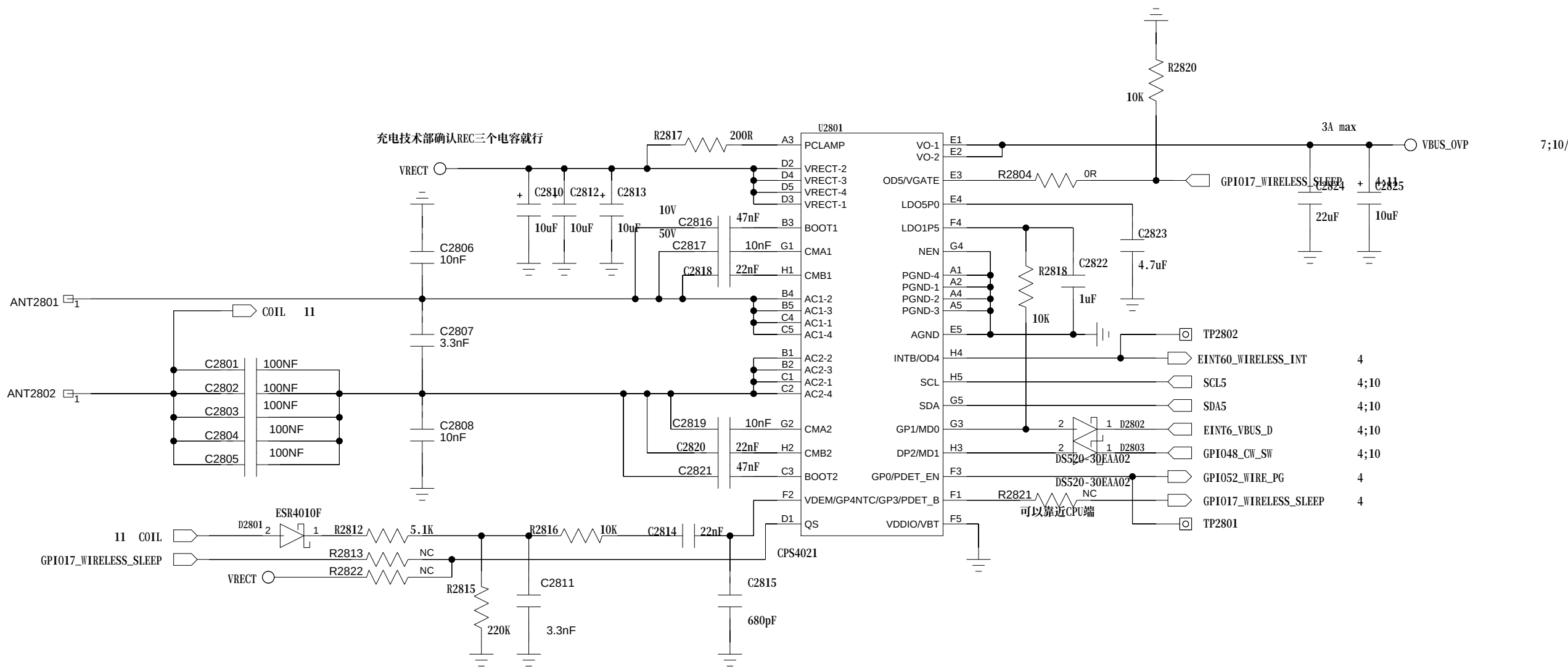
主电池连接器



POWER_THIRD-PARTY-IV

WIRELESS CHG

REVISION RECORD			
LTR	ECO NO:	APPROVED:	DA



	R2813	R2804	R2822	R2821
CPS4021	NC	0R	NC	NC
MT5708	0R	NC	NC	NC
MT5706	NC	NC	0R	0R

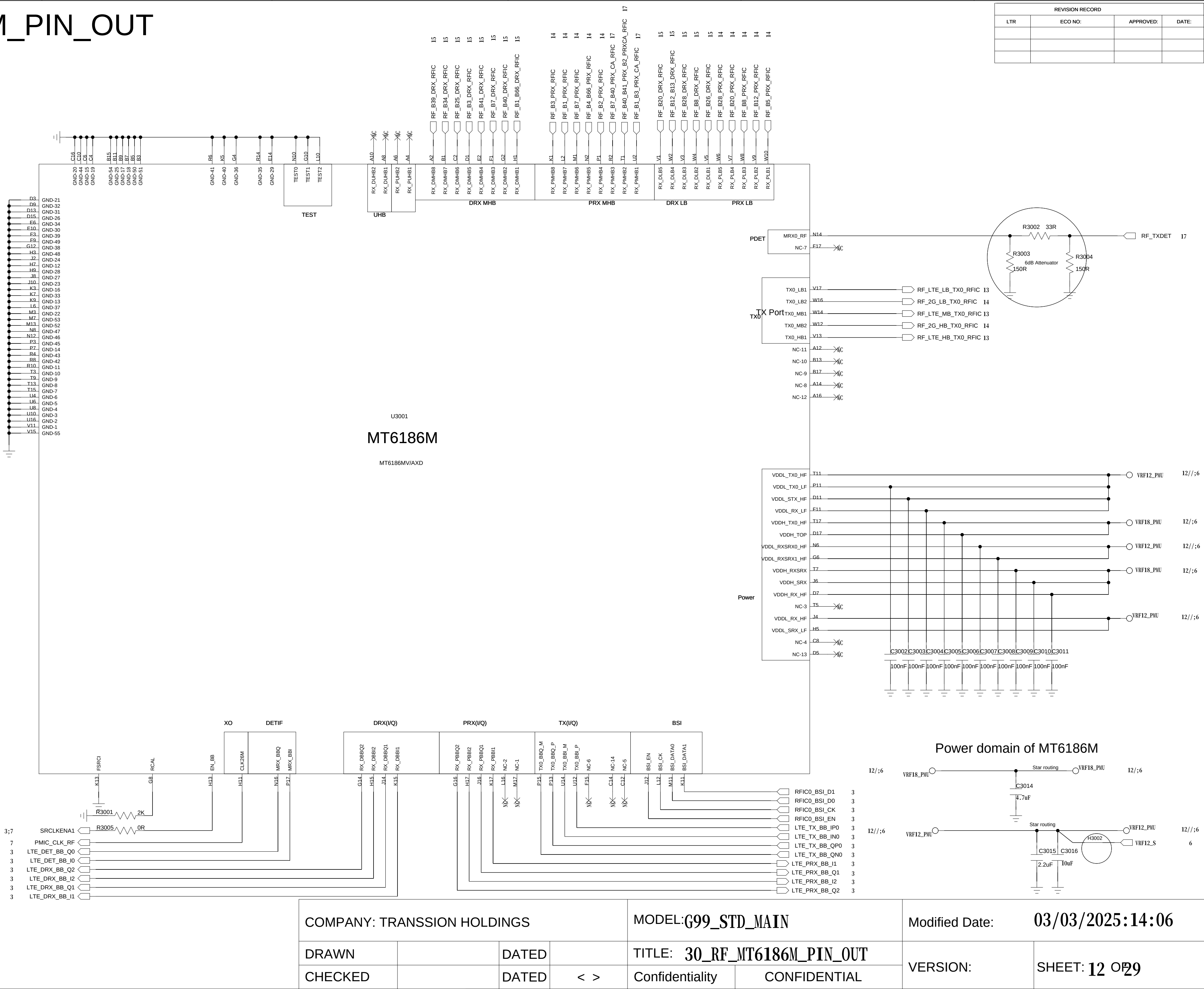
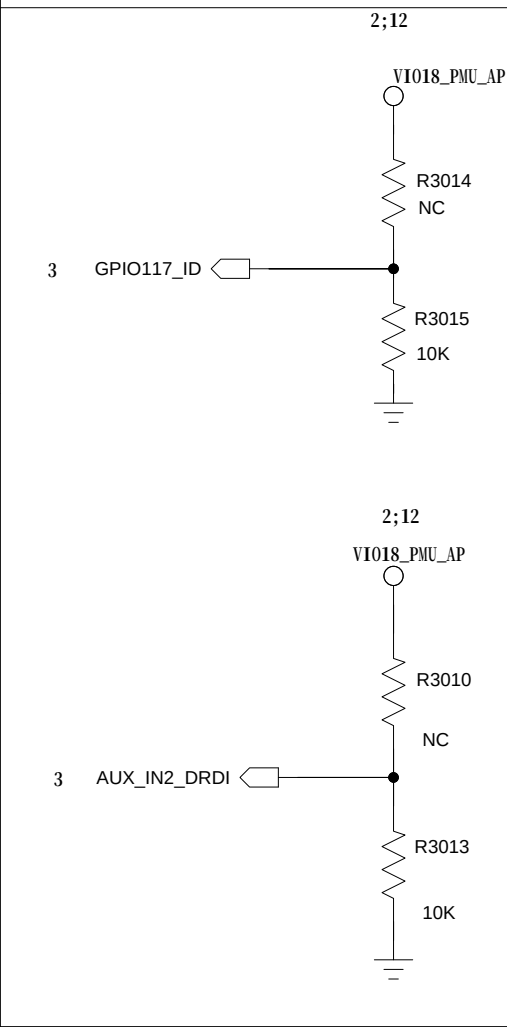
	D1	E3	F1
CPS4021	NC	sleep功能	NC
MT5708	sleep功能	NC	NC
MT5706	VRECT	NC	sleep功能

如果要用MT6375或者MT6360的时候，且PD_VBUS接到OVPIC的前端，不能使用MT5706，MT5708或CPS4021的F1脚接MT6375或MT6360的CHRDETB引脚，

COMPANY: TRANSSION HOLDINGS				MOD 099_STD_MAIN		Modified Date: 03/03/2025:14:06	
DRAWN		DATED		TITLE 28_POWER_THIRD-PARTY-IV		VERSION:	SHEET 11: 20F
CHECKED		DATED	< >	Confidentiality	CONFIDENTIAL		

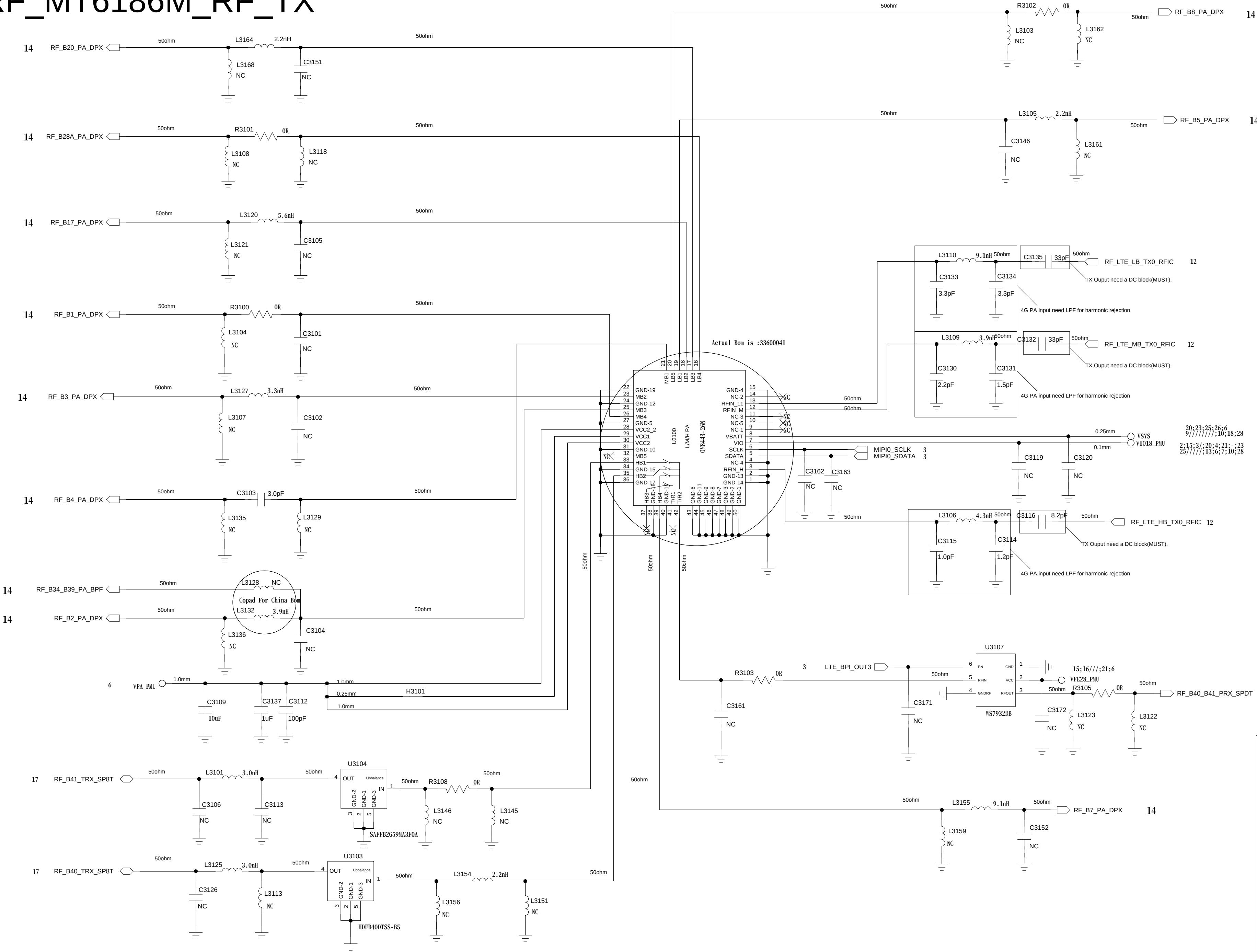
RF_MT6186M_PIN_OUT

REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:



RF_MT6186M_RF_TX

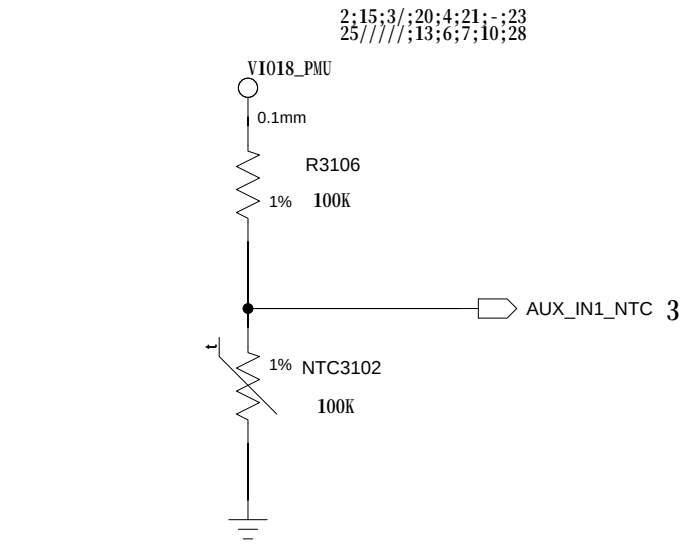
REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:



3/4G_PAIN_LB

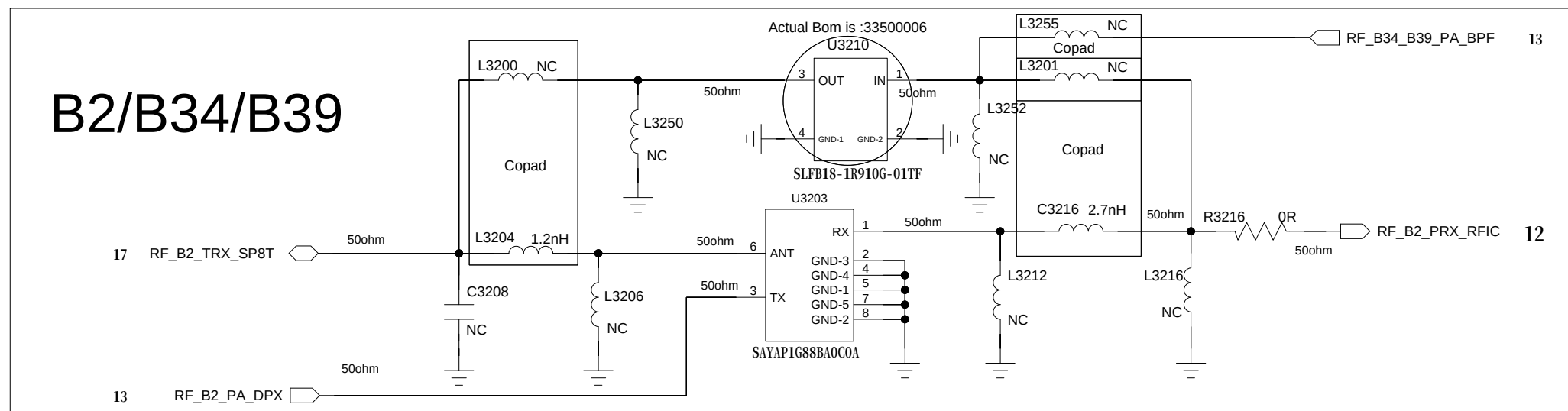
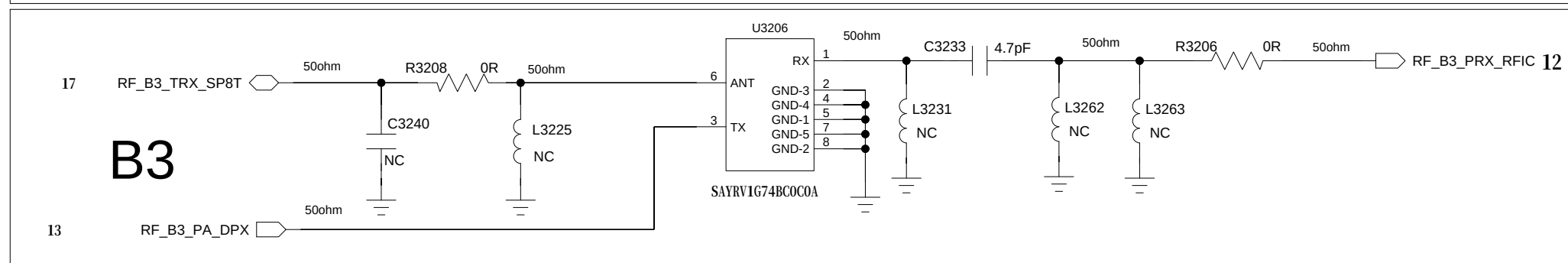
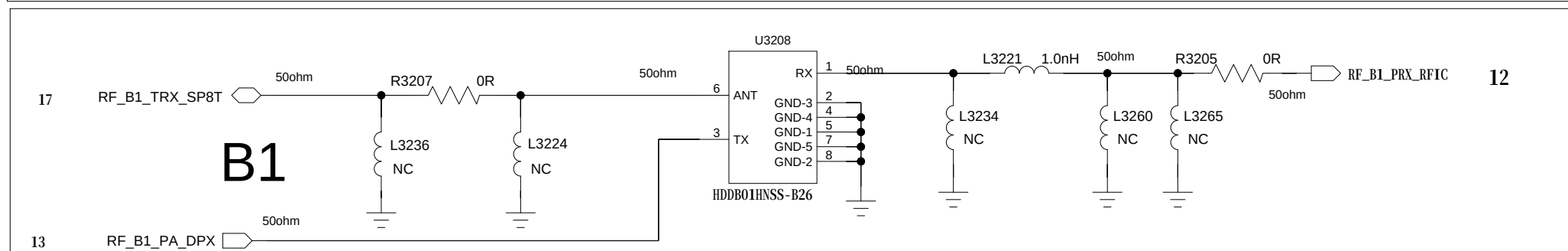
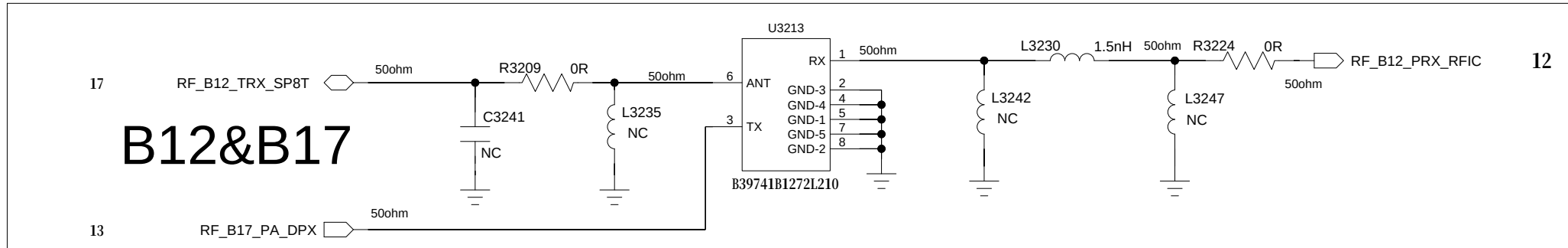
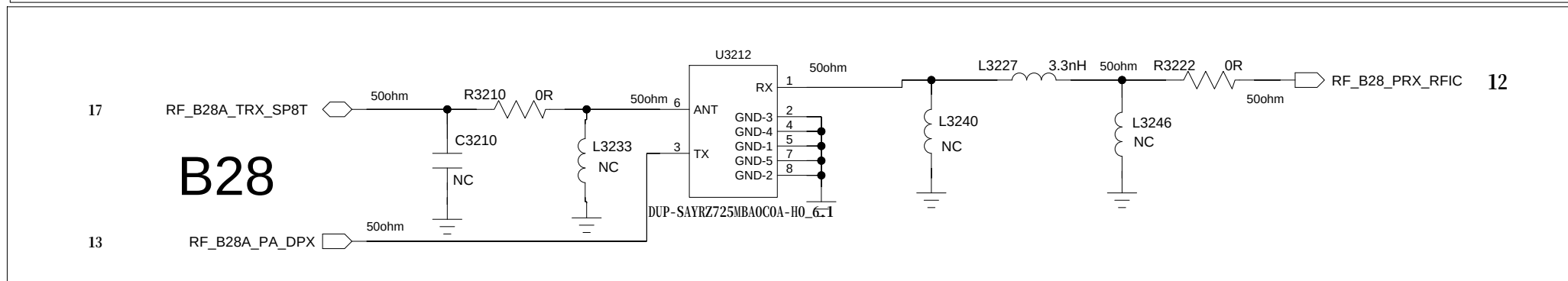
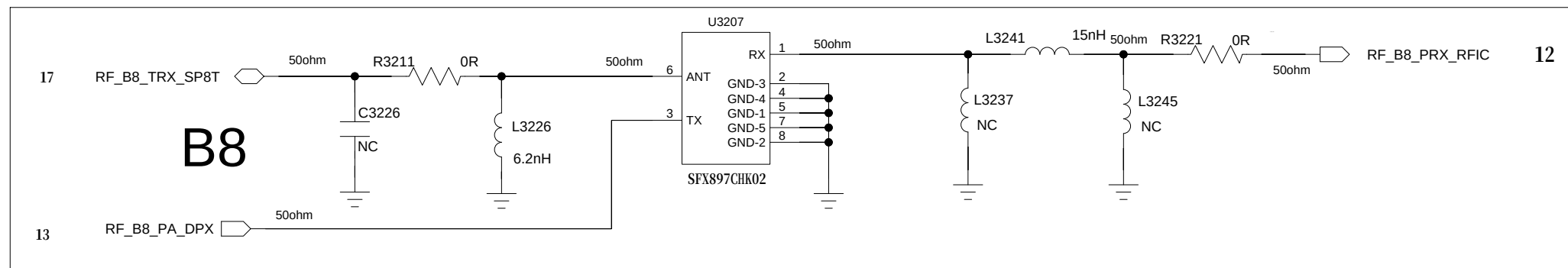
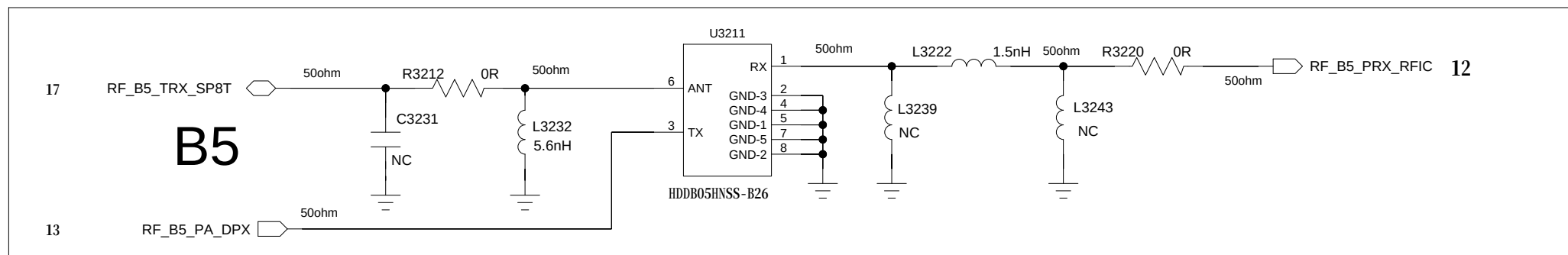
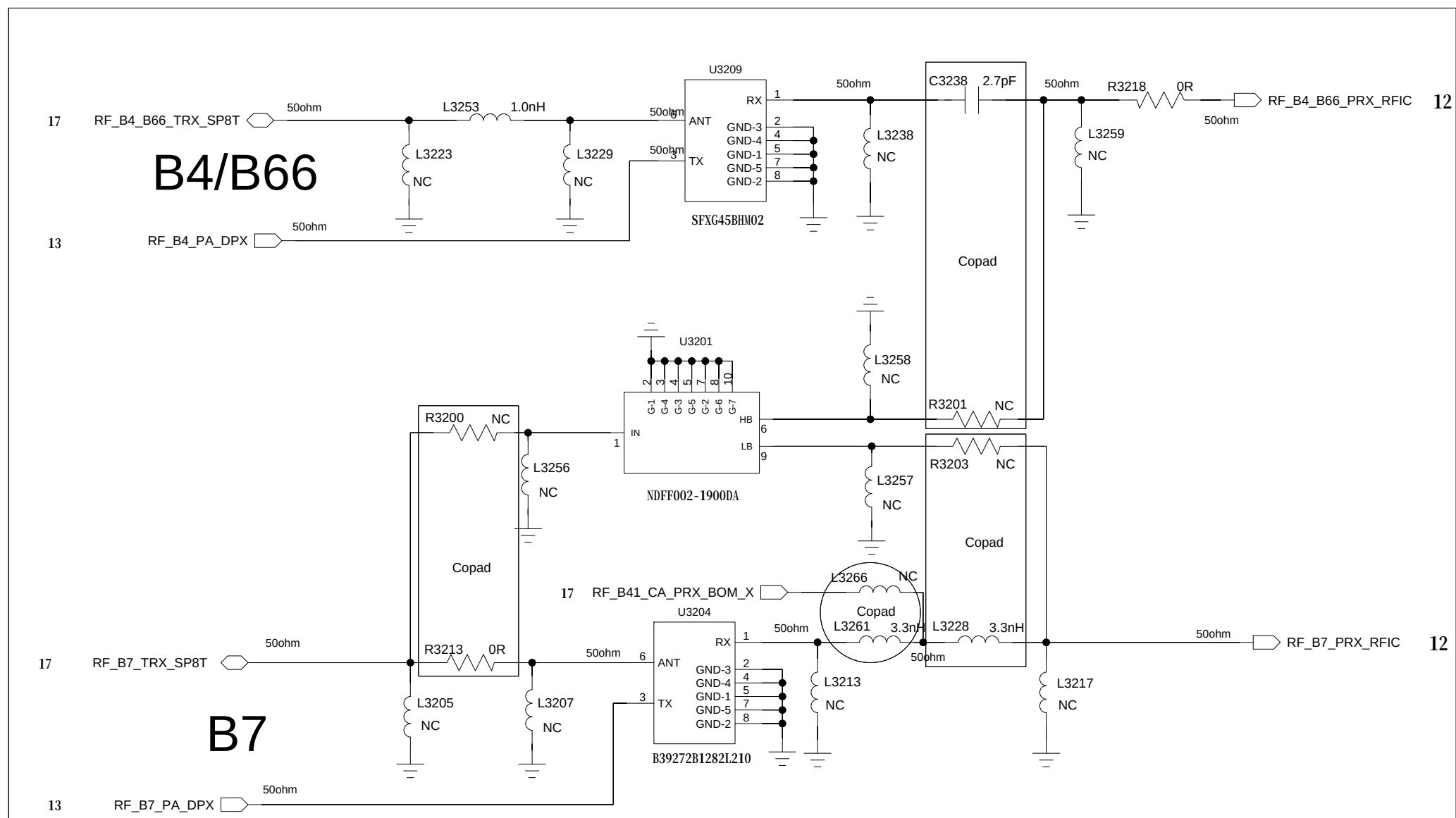
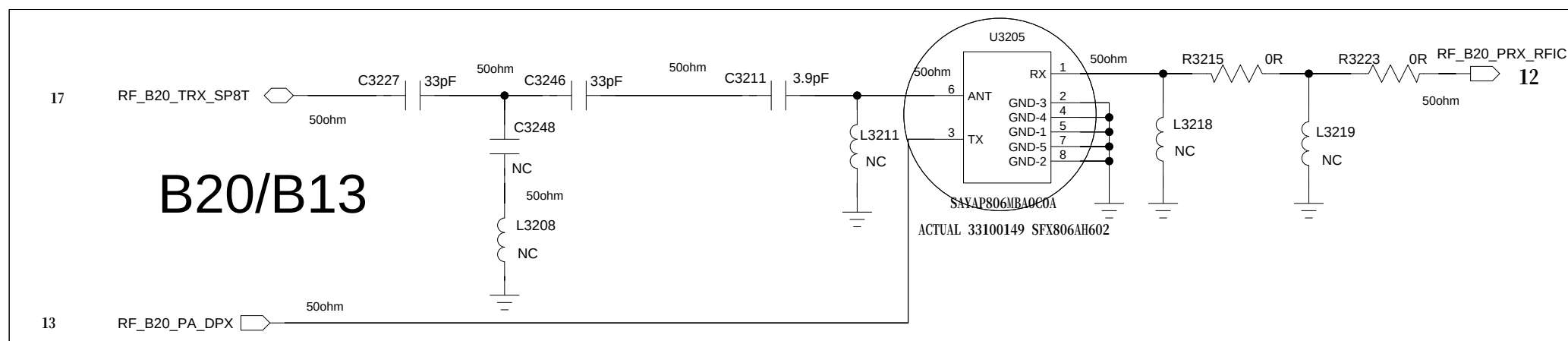
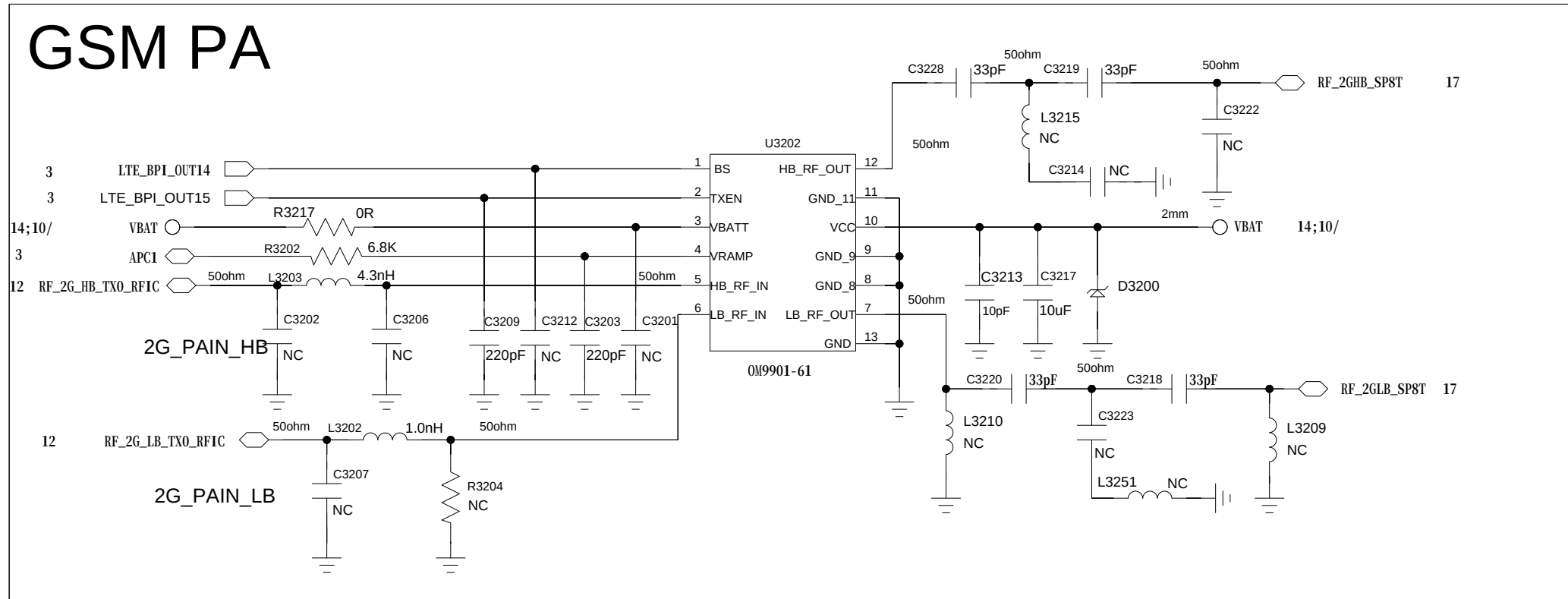
3/4G_PAIN_MB

3/4G_PAIN_HB



COMPANY: TRANSSION HOLDINGS				MODEL:G99_STD_MAIN		Modified Date: 03/03/2025:14:06	
DRAWN		DATED		TITLE: 31_RF_MT6186M_RF_TRX		VERSION:	SHEET: 13 OF 29
CHECKED		DATED	< >	Confidentiality	CONFIDENTIAL		

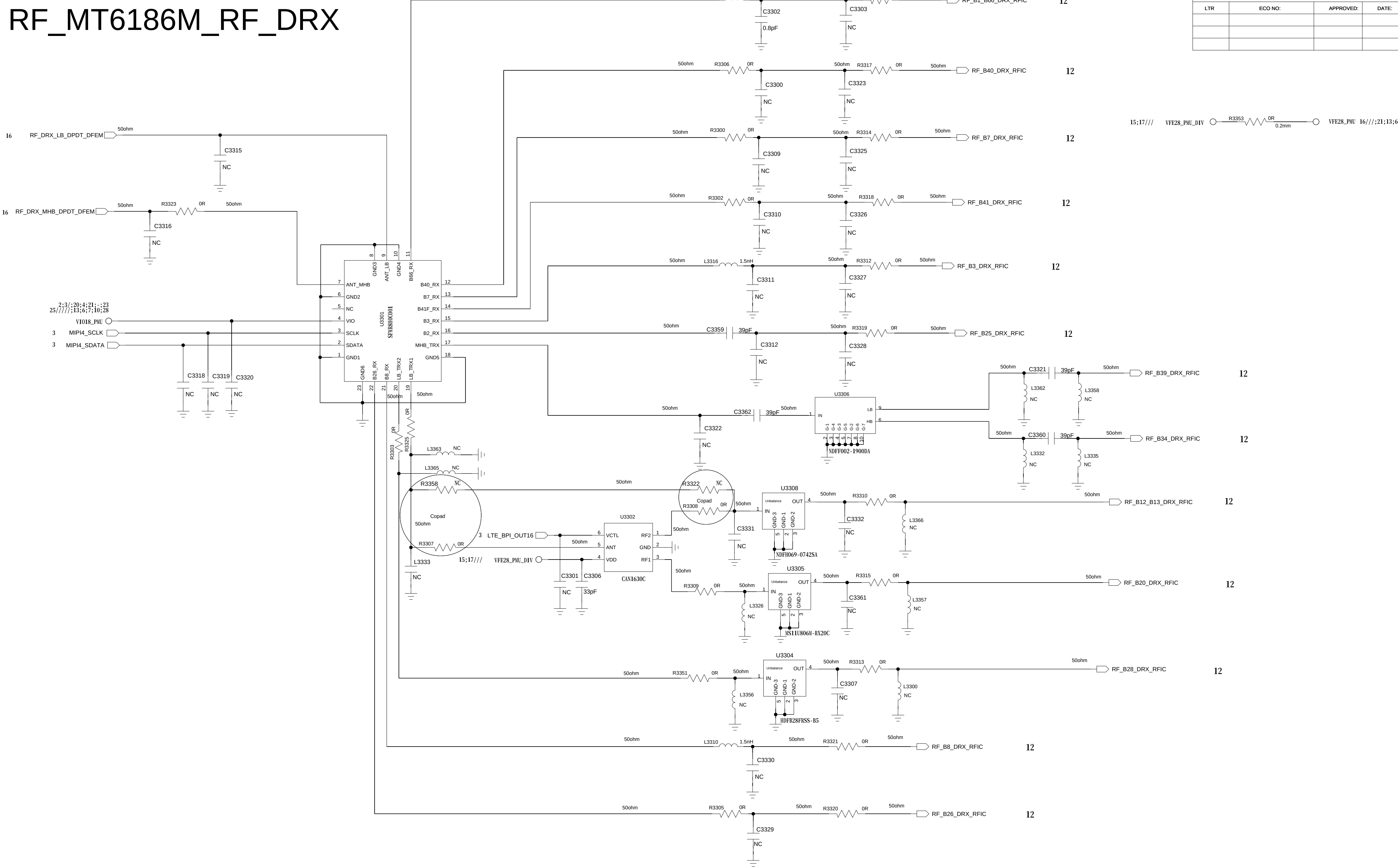
GSM PA



COMPANY: TRANSSION HOLDINGS				MODEL:G99_STD_MAIN		Modified Date: 03/03/2025:14:06	
DRAWN		DATED		TITLE: 32_RF_MT6186M_RF_PRX		VERSION:	SHEET: 14 of 29
CHECKED		DATED	< >	Confidentiality	CONFIDENTIAL		

RF_MT6186M_RF_DRX

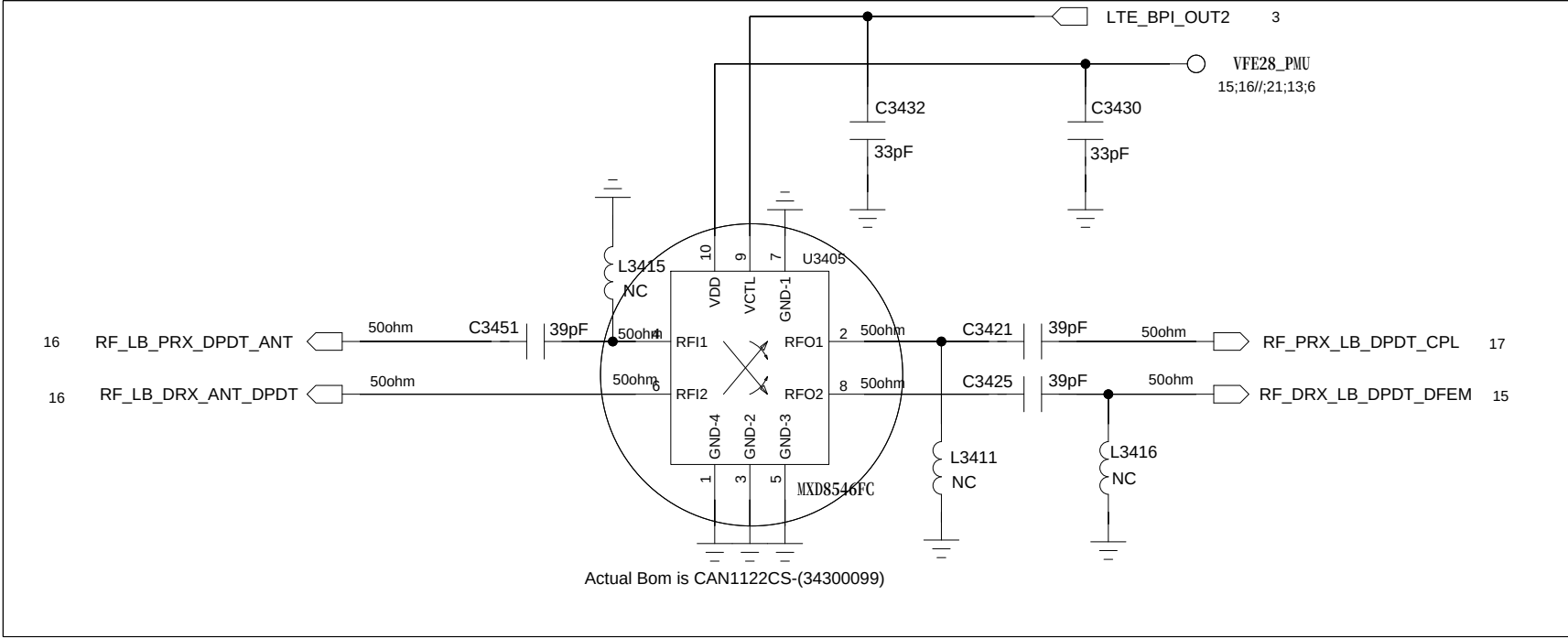
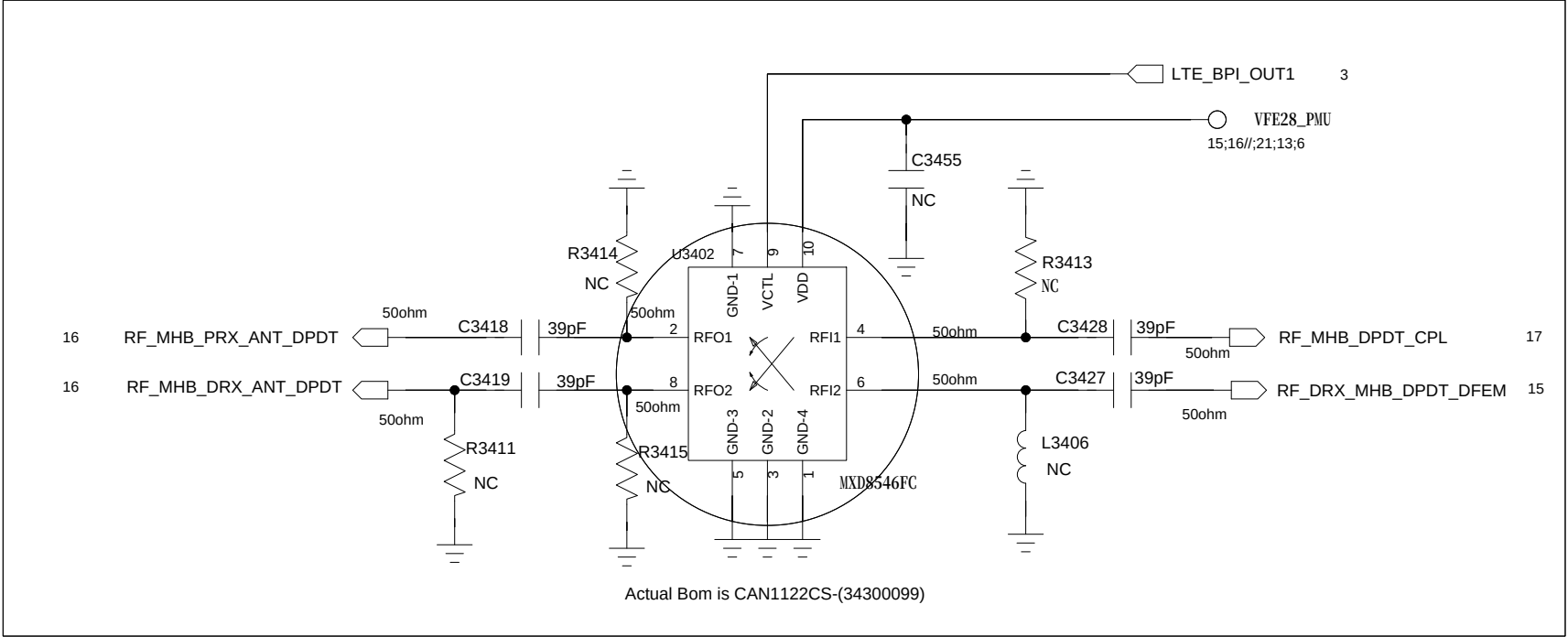
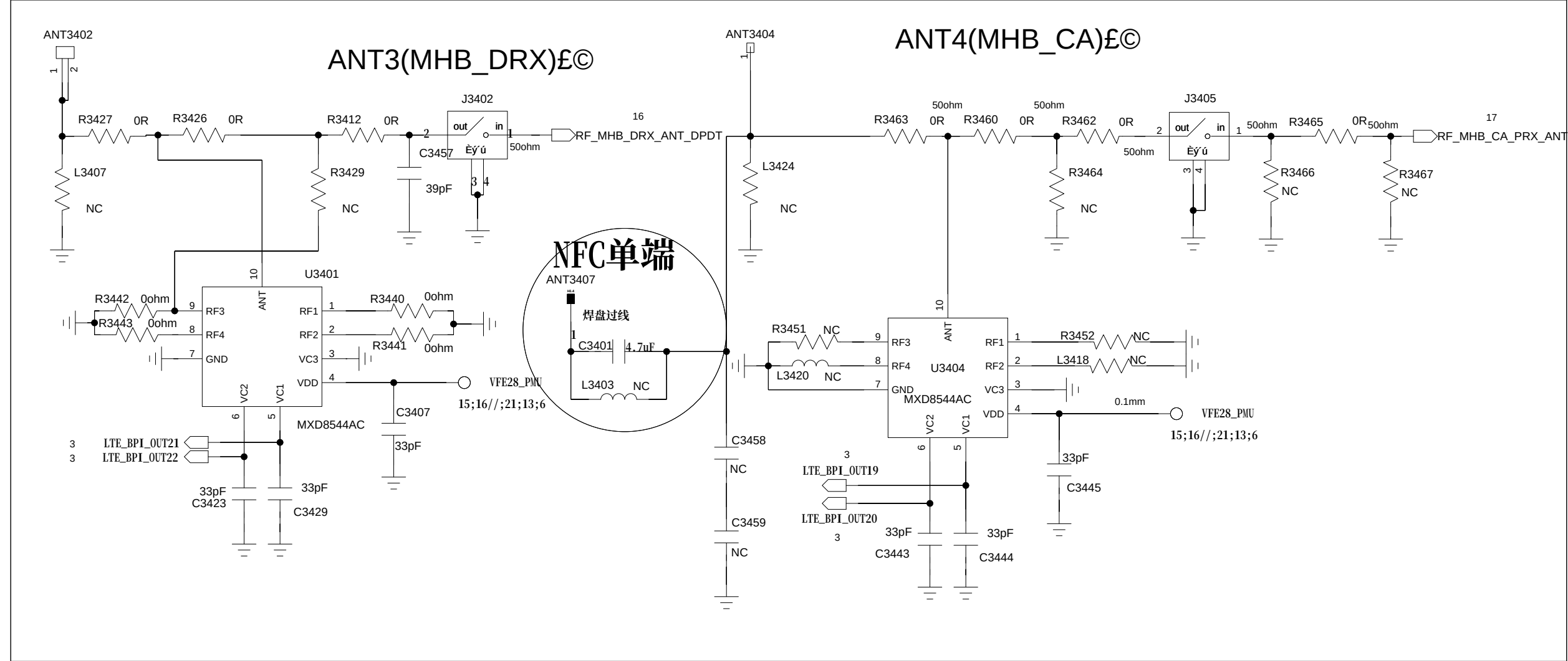
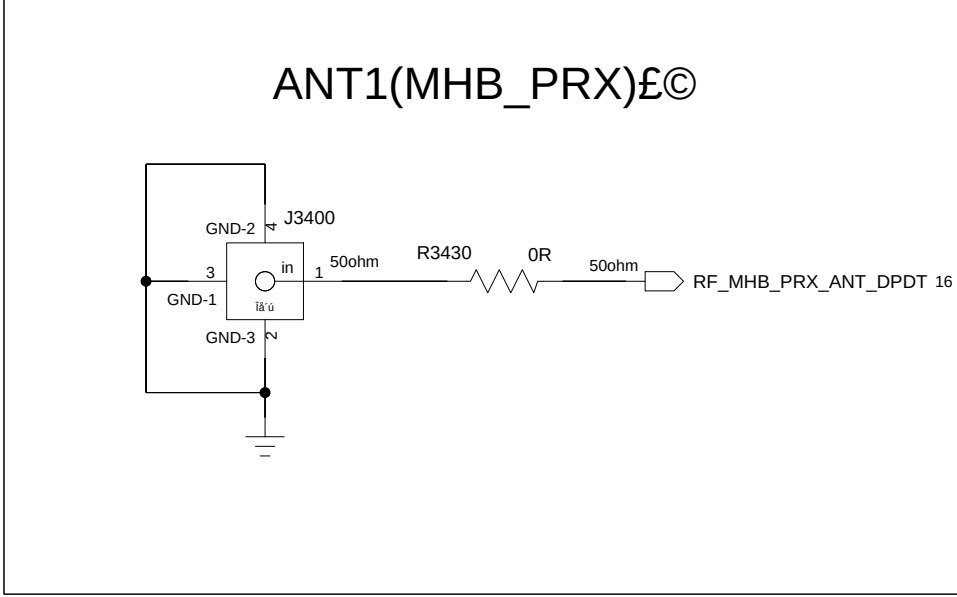
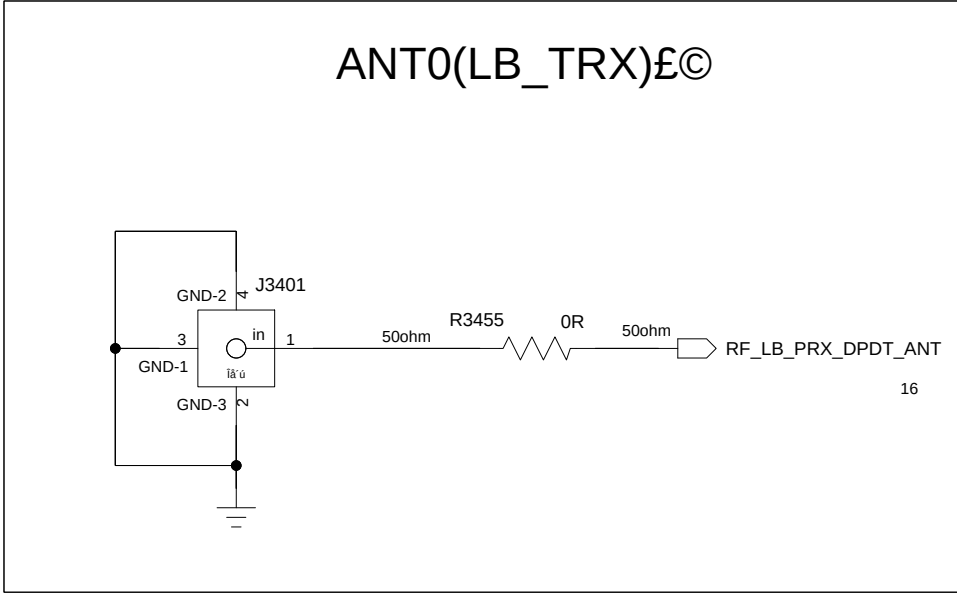
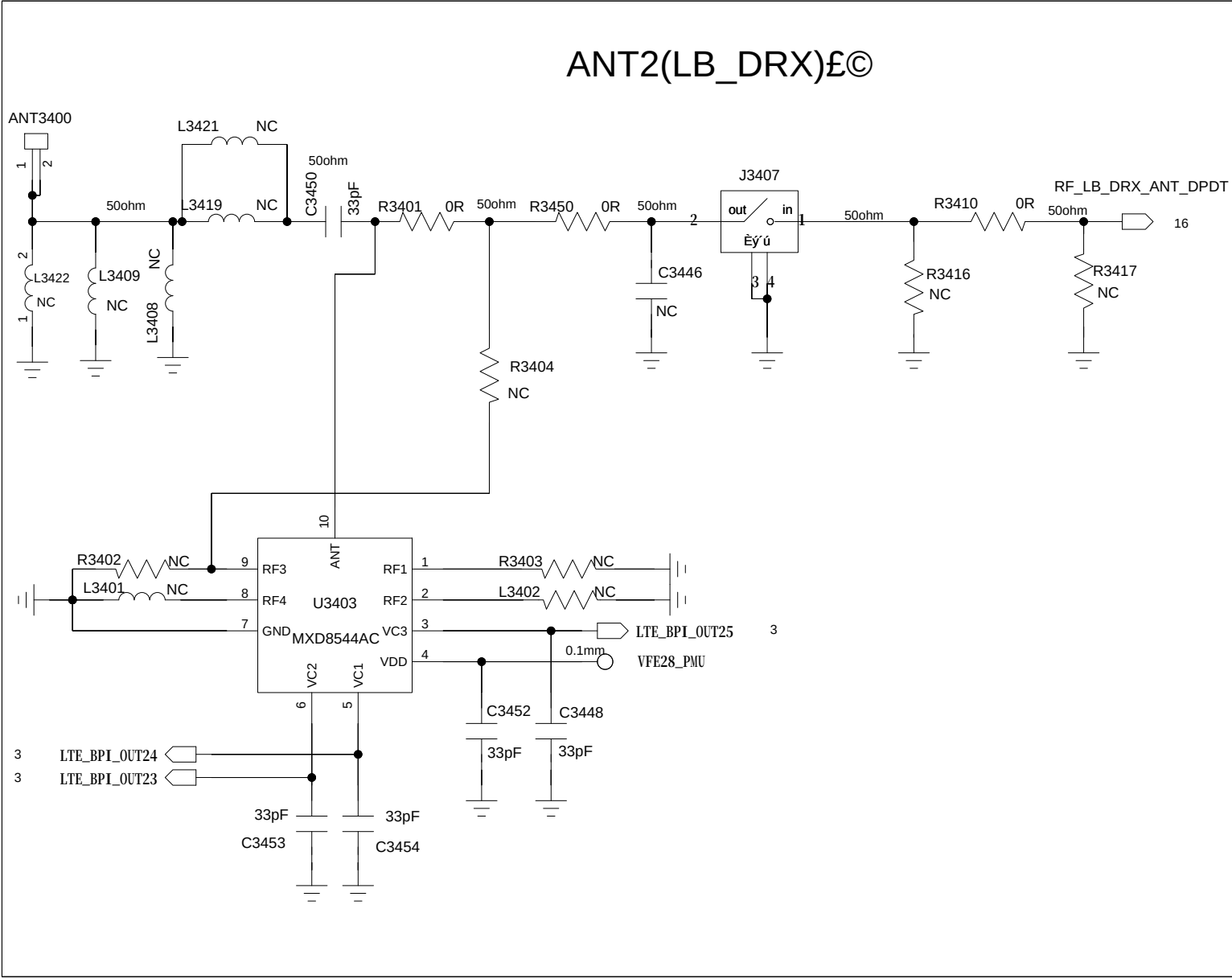
REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:



COMPANY: TRANSSION HOLDINGS				MODEL:G99_STD_MAIN		Modified Date:	03/03/2025:14:06
DRAWN		DATED		TITLE: 33_RF_MT6186M_RF_DRX		VERSION:	SHEET: 15 OF 29
CHECKED		DATED	< >	Confidentiality	CONFIDENTIAL		

RF_MT6186M_RF_ANT

REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:



Ground Spring

screw Spring

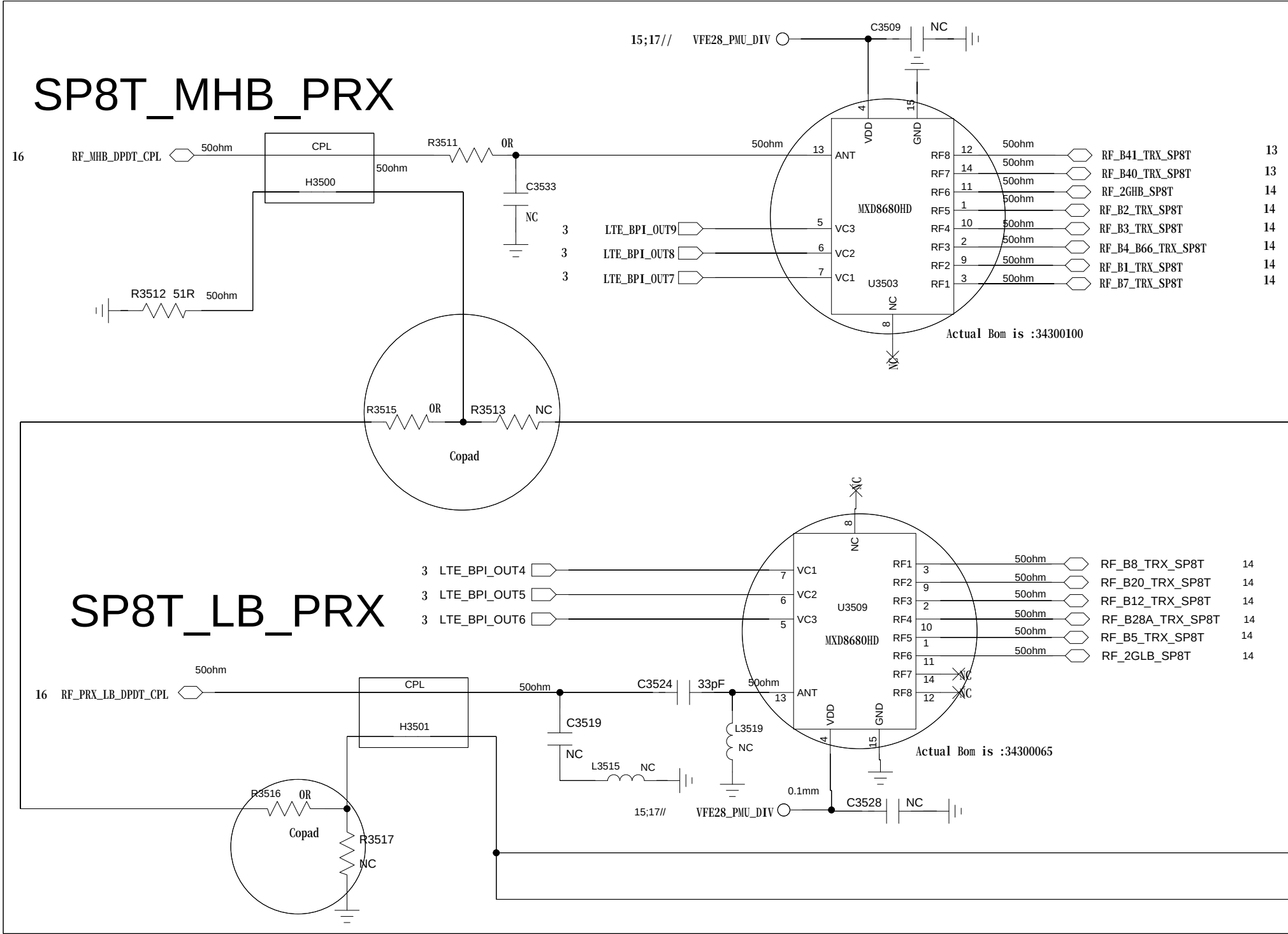
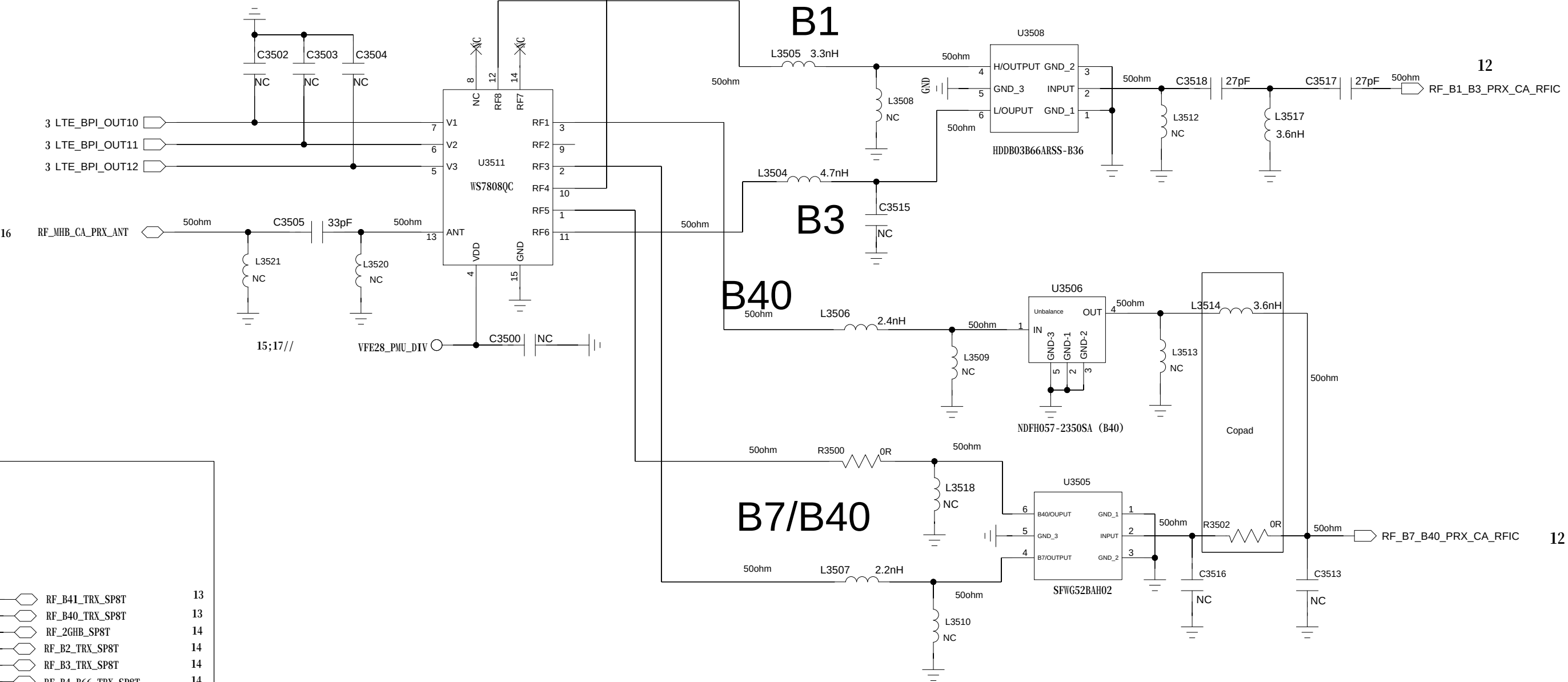
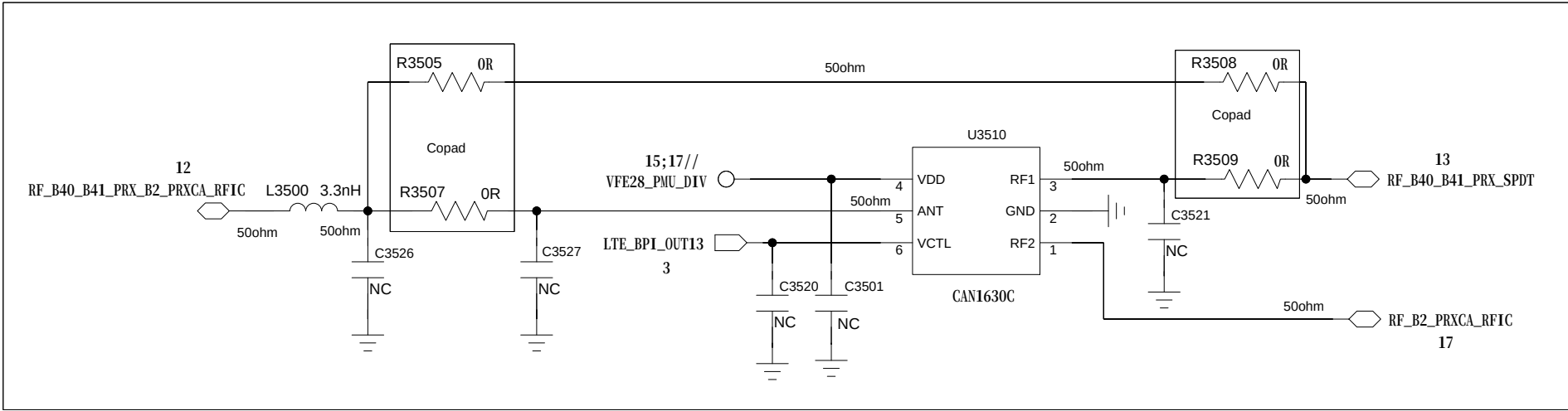
top Spring

bot Spring

COMPANY: TRANSSION HOLDINGS				MODEL: G99_STD_MAIN		Modified Date: 04/03/2025:10:34	
DRAWN		DATED		TITLE: 34_RF_ANT_CONTROLLER		VERSION:	SHEET: 16 of 29
CHECKED		DATED	< >	Confidentiality	CONFIDENTIAL		

RF_MT6186M_RF_CA

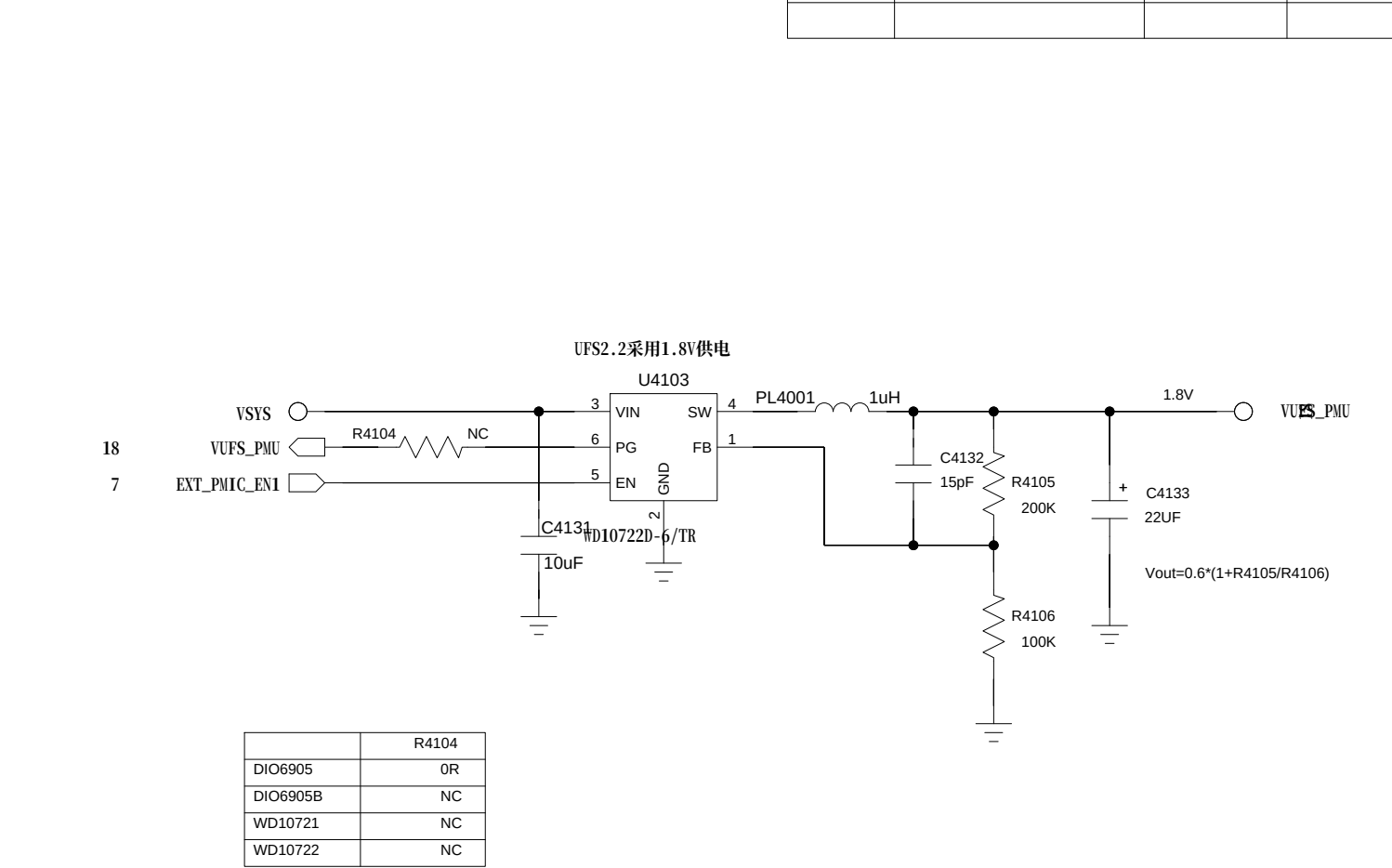
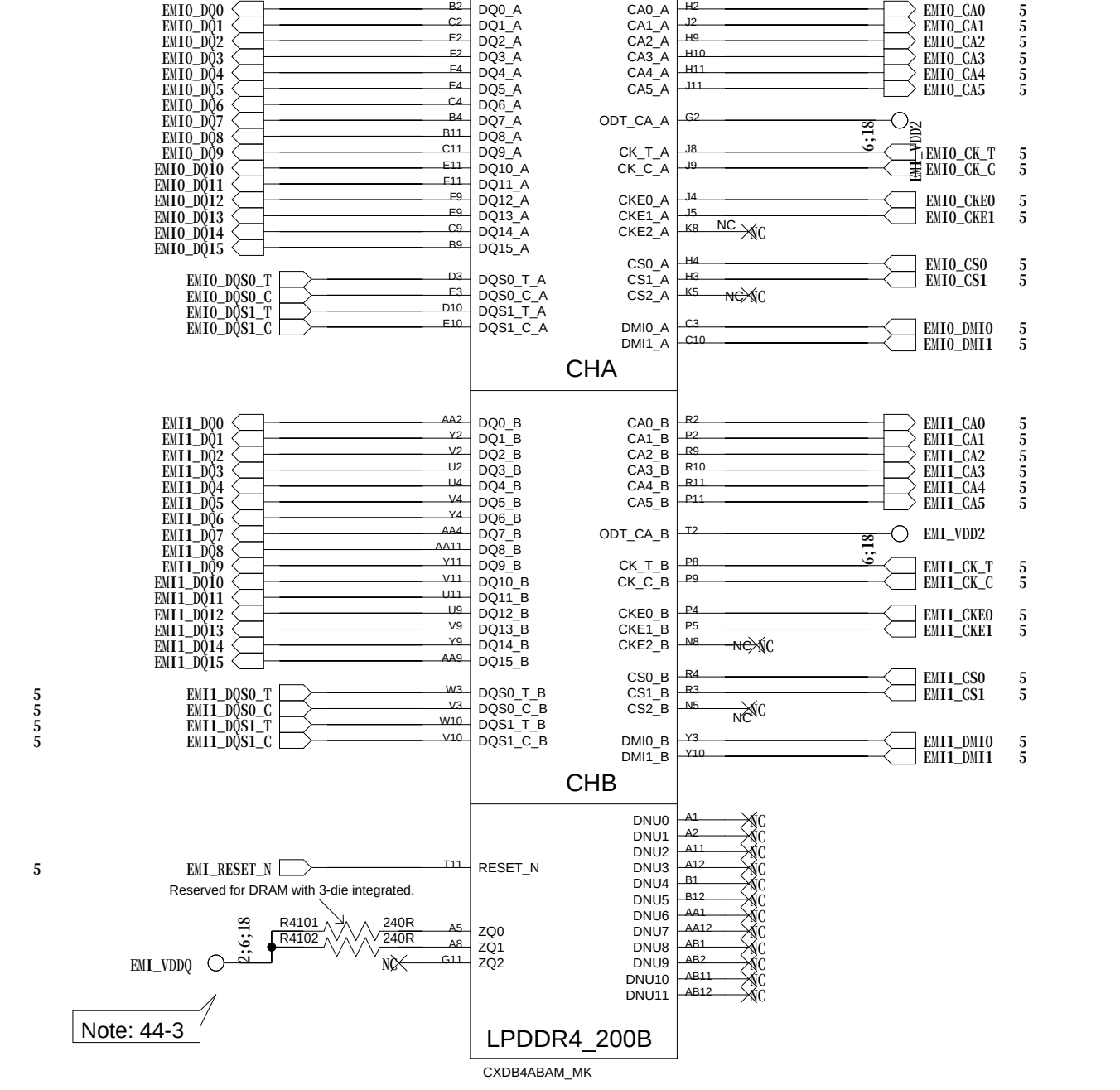
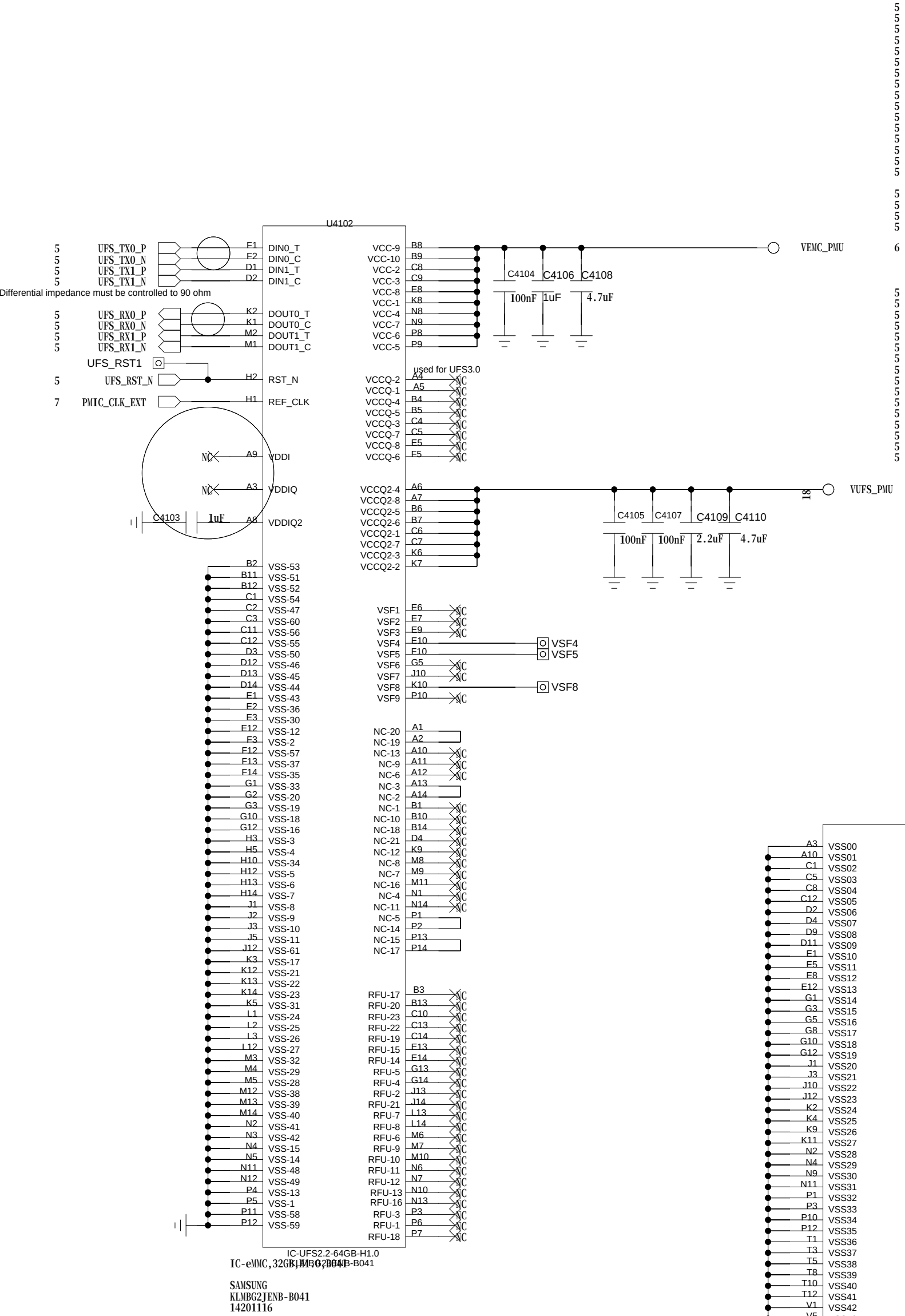
REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:



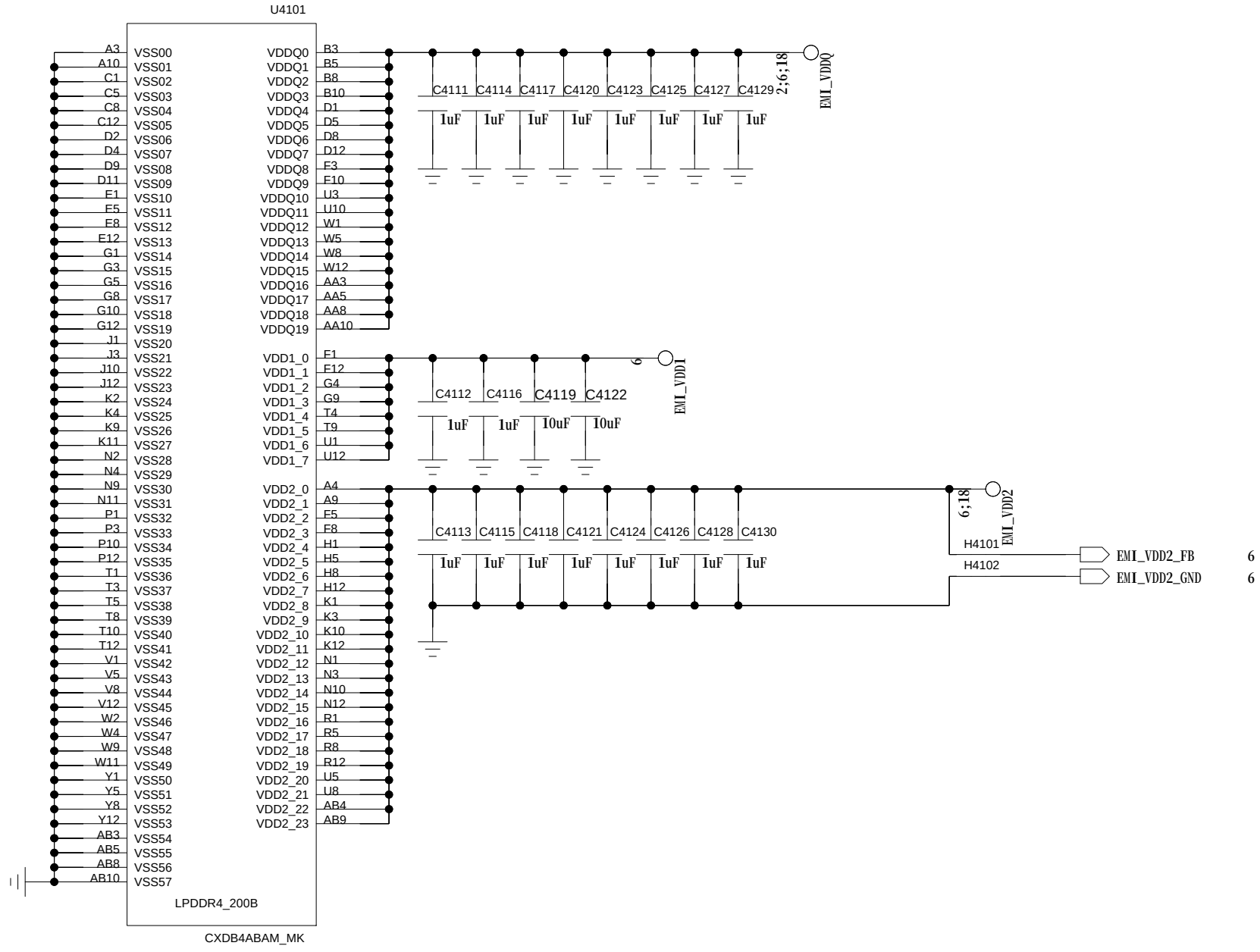
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DRAWN		DATED		TITLE: 35_RF_MT6186M_RF_MHB_CA		VERSION:	SHEET: 17 OF 29
CHECKED		DATED	< >	Confidentiality	CONFIDENTIAL		

MEM_DSC_LPDDR4x

UFS3.1的VEMC采用2.55V供电，切换芯片注意更换FB电阻搭配，不同IC有细微差异



Note: 44-3



COMPANY: TRANSSION HOLDINGS

MODEL:G99_STD_MAIN

Modified Date: 03/03/2025:14:06

DRAWN

DATED

TITLE: 41_MEMORY_DSC_LPDDR4X

CHECKED

DATED

< >

Confidentiality

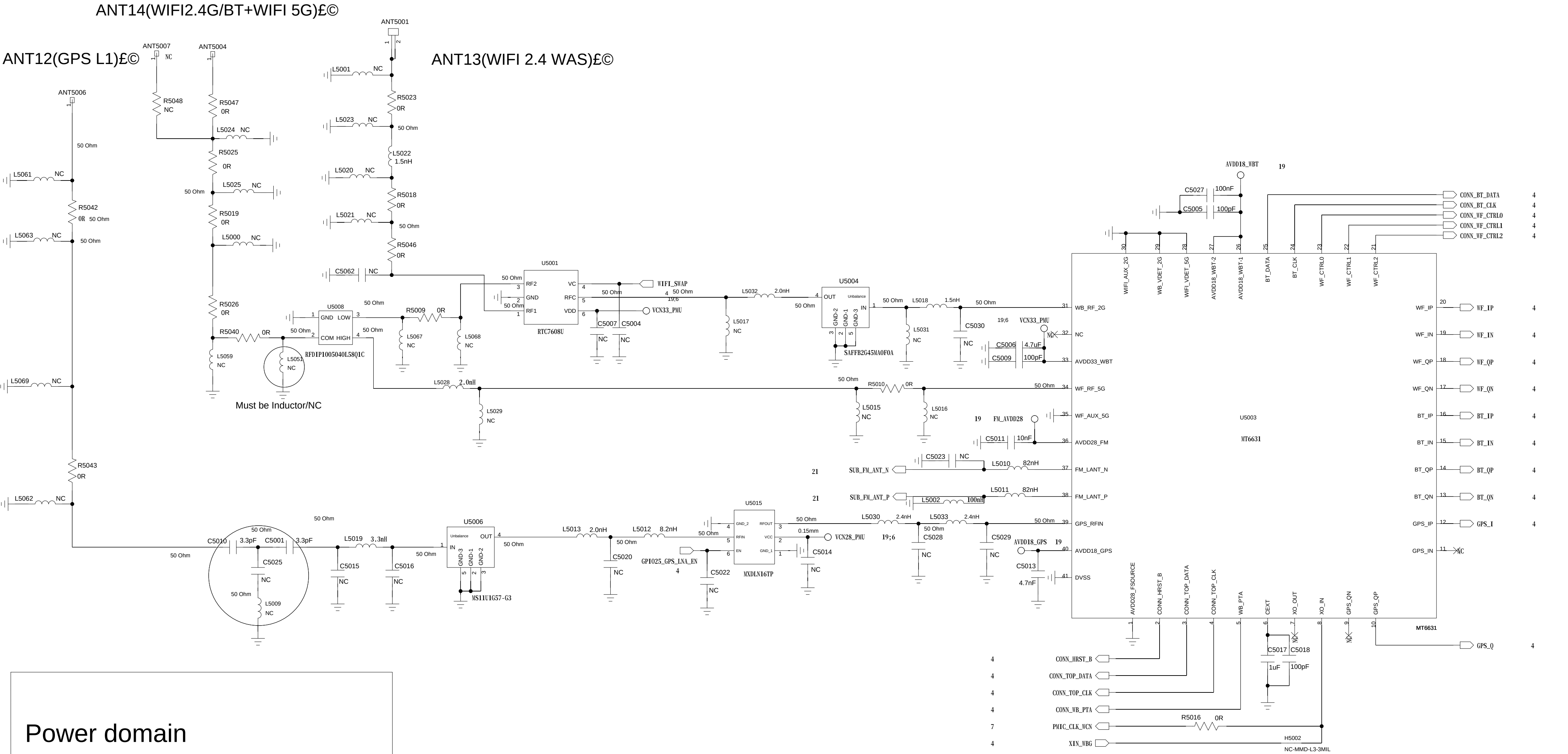
CONFIDENTIAL

VERSION:

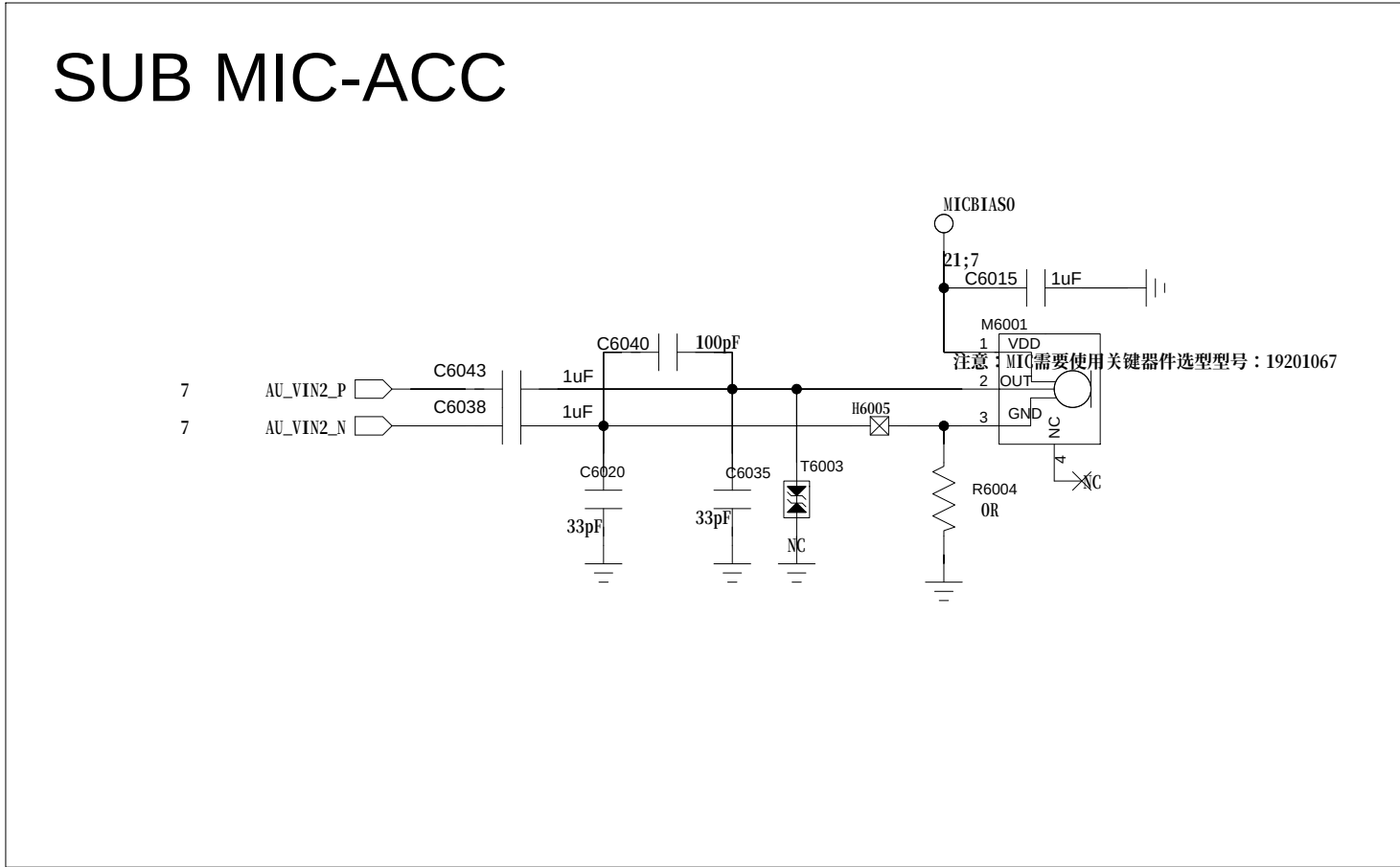
SHEET: 18 OF 29

CONNECTIVITY_MT6631

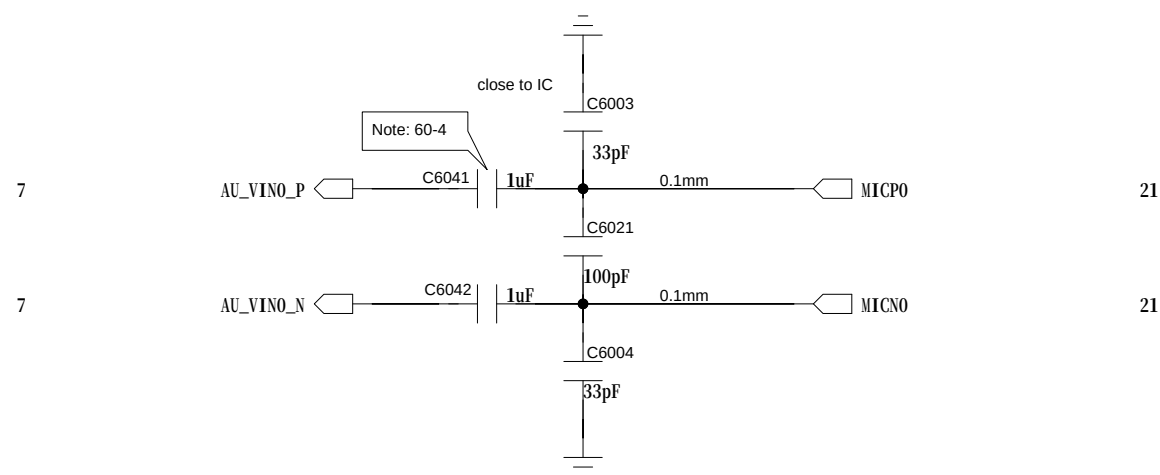
REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:



Audio PA FS1599N:I2C ADDRESS:0x6A(W)/0x6B(R)
UP SPK AW87390FCR:I2C A1=VI018 ADDRESS:0xB2(W)/0xB3(R)
AW87390,0XB2/B3



MAIN MIC



For multi-key accuracy at DCC mode
 Default use internal PU-Resistor; for one key solution
 Use external-Rresistor(R6015);for multi-key solution
 Detail SW compile option setting,please refer design note

Note: 60-2

Note: 60-1

0.15mm

AU_HPR

AU_HPL

R6012 20R

R6013 20R

close to IC

XMP3_

XMP3_

C6023 33pF

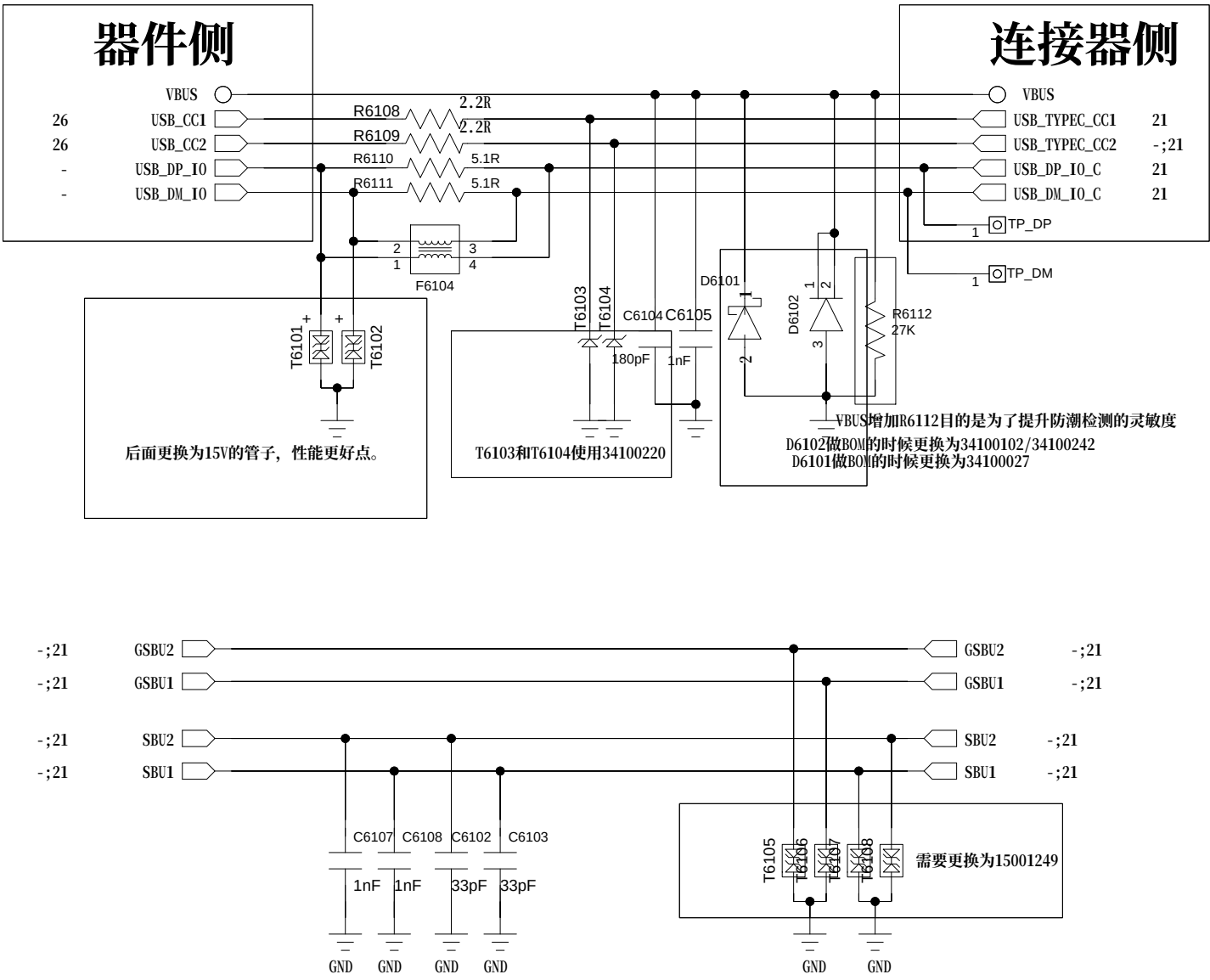
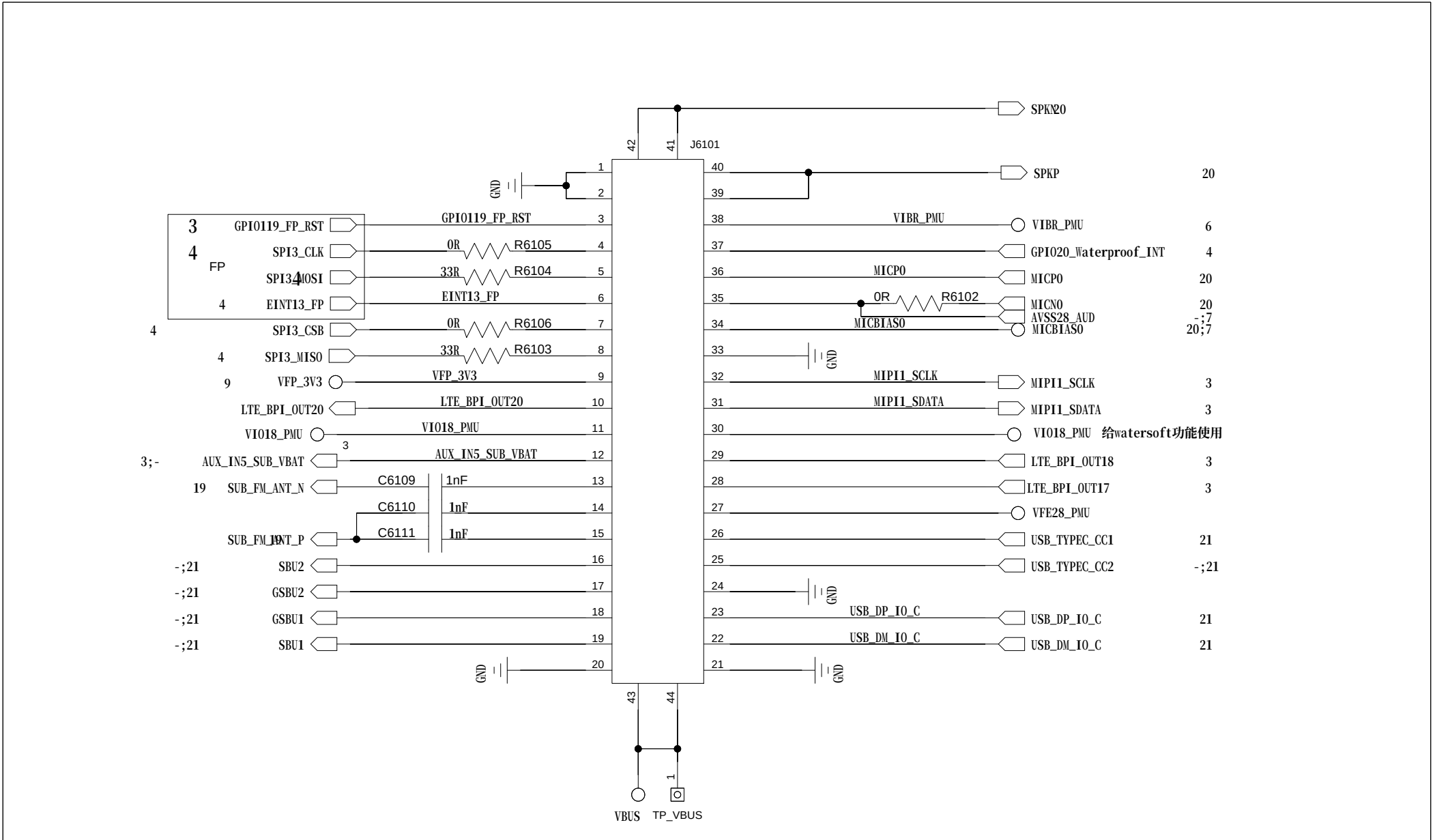
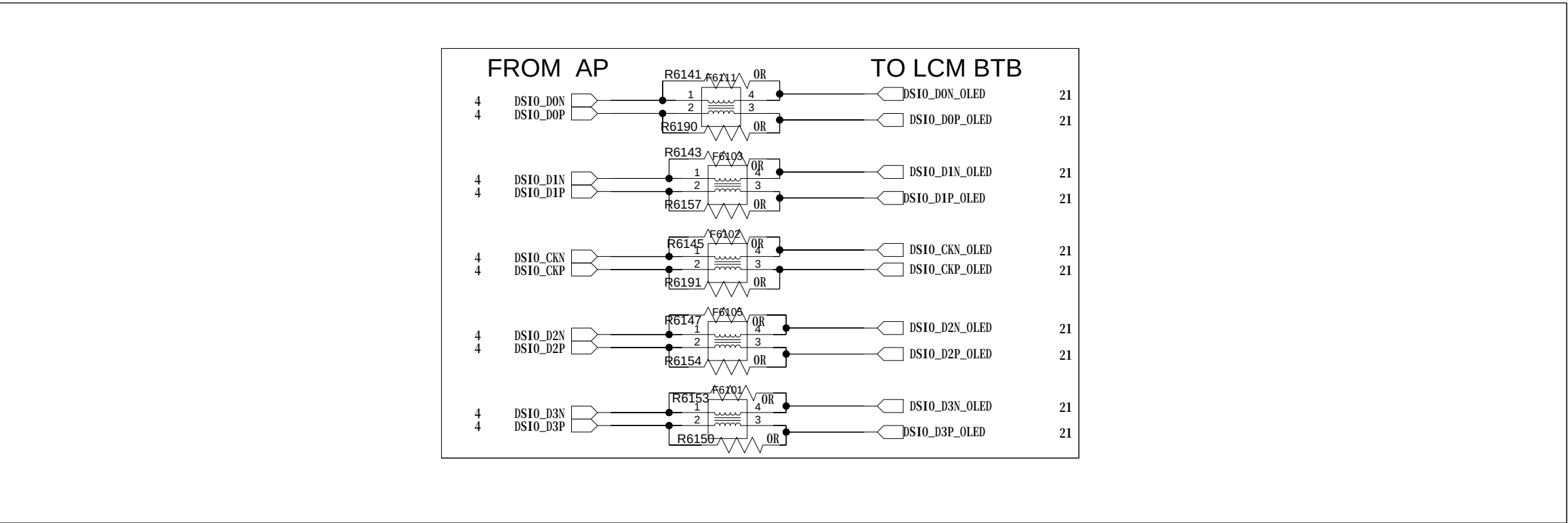
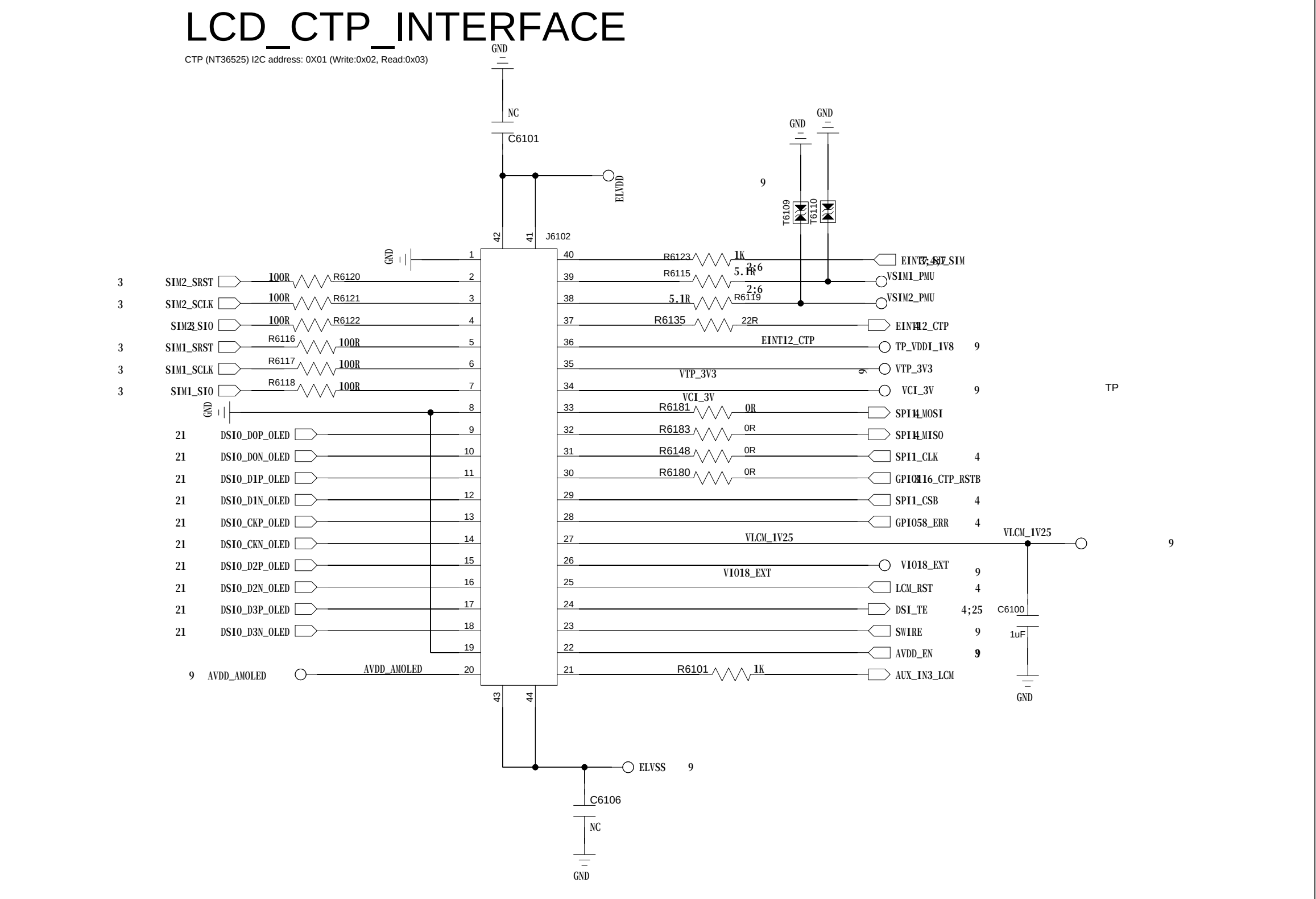
C6029 33pF

Based on your system level design , if better Audio performance is needed on your system, please add 32ohm to audio path for performance enhance proposal (32ohm condition pop noise can improve 6dB)

Schematic design notice of "60_PERI_AUDIO_IO" page.	
Note 60-1:	B6009 B6010 B6013 B6014 needs change to "BLM18BD102SN1" for high THD performance(-90dB), but this BOM change will result in FM RSSI 10dB degraded.
Note 60-2:	To reserve a resistor in HPL and HPR in series connection both in order to optimize headphone pop noise. The recommended value of this resistor is 33R.
Note 60-3:	Layout trace from MT6353 pin J3 AUDREFN to Audio jack GND must surround shield with GND.
Note 60-4:	0.1/1uF for ACC mode(1uF for WB_AMR Speech/0.1uF for NB_AMR Speech),0R for DCC mode.

COMPANY: TRANSSION HOLDINGS				MODEL:G99_STD_MAIN		Modified Date: 03/03/2025:14:06	
DRAWN		DATED		TITLE: 60_PERI_AUDIO		VERSION:	SHEET: 20 OF 29
CHECKED		DATED	< >	Confidentiality	CONFIDENTIAL		

PERI_LCM_CTP_FP

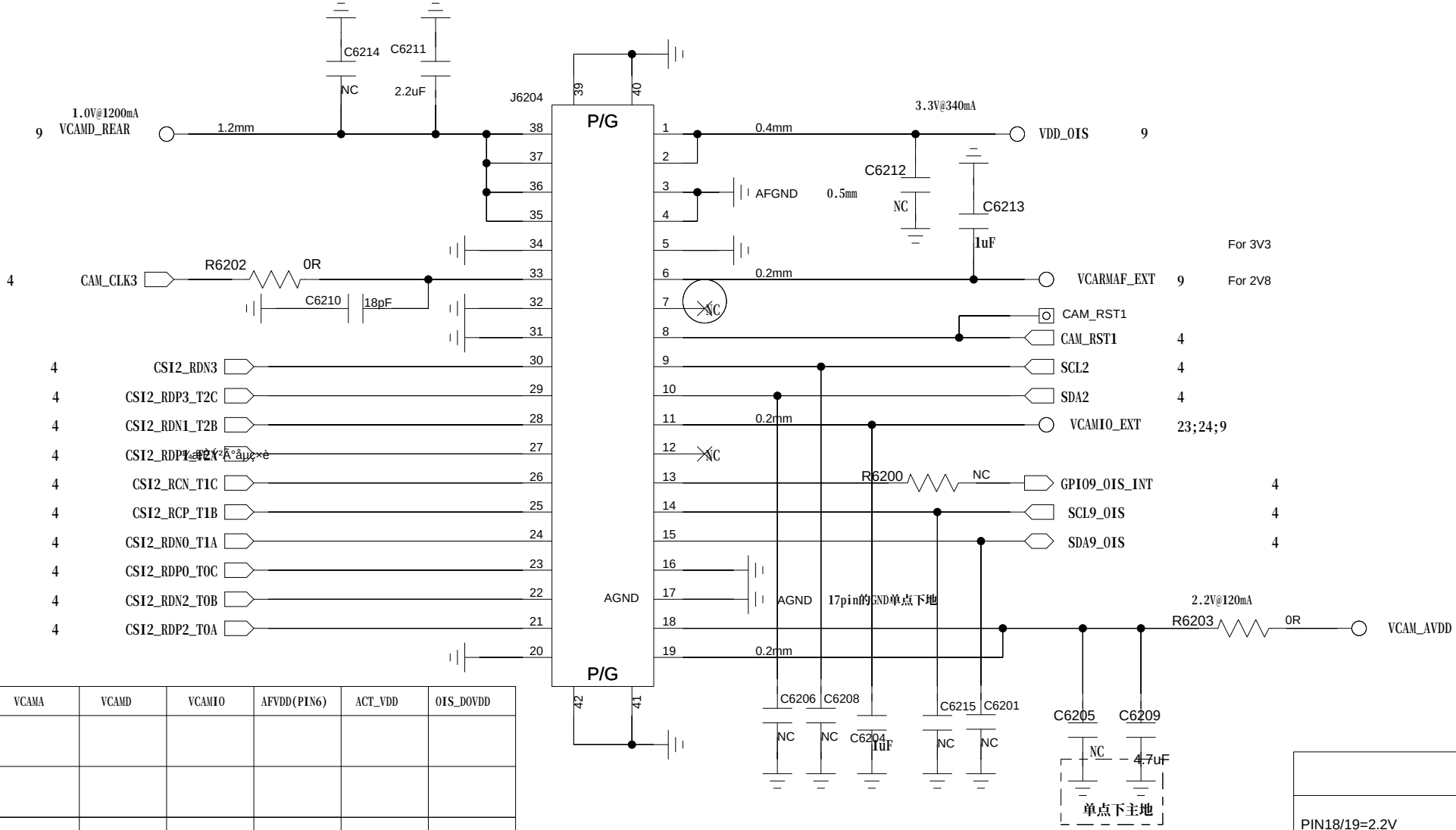


COMPANY: TRANSSION HOLDINGS				MODEL:G99_STD_MAIN		Modified Date:	04/03/2025:22:10
DRAWN		DATED		TITLE: 61_PERI_LCM_CTP_FP		VERSION:	
CHECKED		DATED	< >	Confidentiality	CONFIDENTIAL	SHEET:	21 OF 29

PERI_CAMERA_I

RCAM 50M+2M

REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:



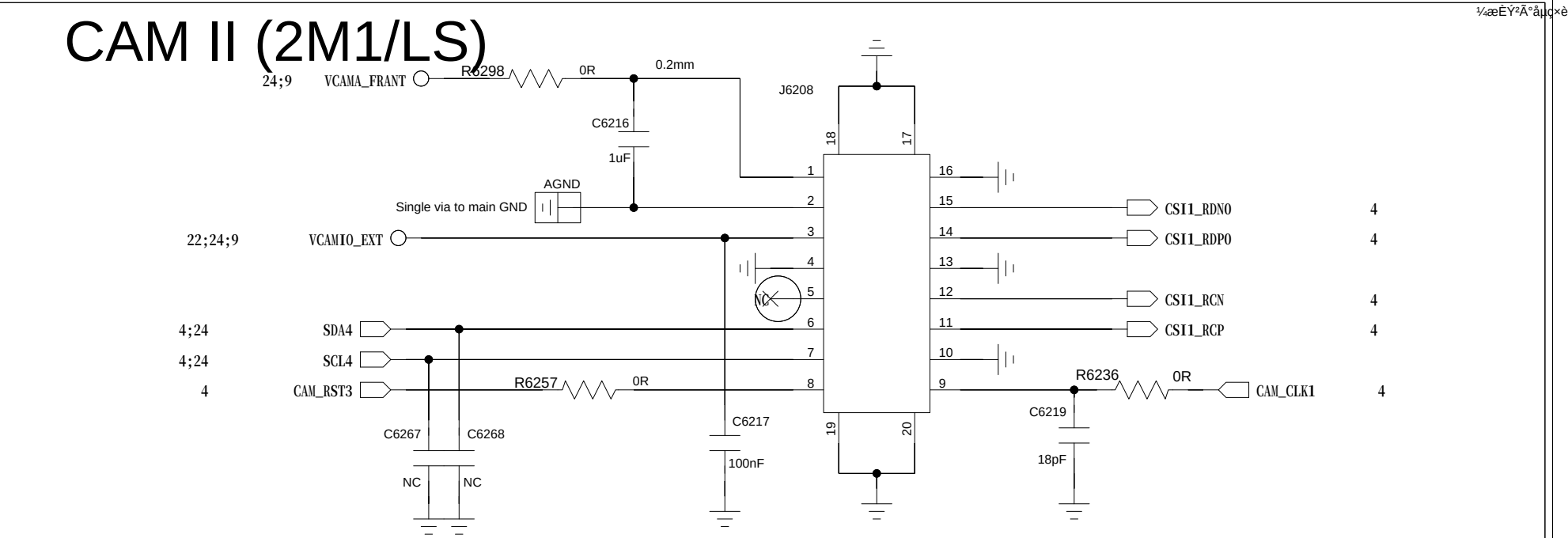
9

	R6230	R6235	
PIN18/19=2.2V	0R	NC	SMT 4CH-LDO U2606-2.2v
PIN18=2.9V	NC	0R	SMT 4CH-LDO U2606-2.9v
PIN19=1.8V			SMT U2416-1.8VLDO

SAP	SENSOR	I2C Address	VCAMA	VCAMD	VCAM10	AFVDD(PIN6)	ACT_VDD	OIS_DOVDD
50W 0.81F	IX882-AJB5-C	Write:0x20 Read:0x21	2.8V@ 78.3mA	1.1V@417.2mA	1.8V @3.9mA	2.8V @105mA	NC NC	NC NC
50W 1.04F	GC5082-A12A	Write:0x20 Read:0x21	2.8V@ TBD	1.1V@TBD	1.8V @TBD	2.8V @105mA	NC	NC

COMPANY: TRANSSION HOLDINGS				MODEL:G99_STD_MAIN		Modified Date: 03/03/2025:14:06	
DRAWN		DATED		TITLE: 62_PERI_CAMERA_I		VERSION:	SHEET: 22 OF 29
CHECKED		DATED	< >	Confidentiality	CONFIDENTIAL		

CAM II (2M1/LS)



sensor:0x20(W)/0x21(R)						
	SENSOR	I2C Address	VCAMA	VCAMD	VCAMIO	AFVDD
AI 17201873	MS-P0T17-31C-L4G1-T88	NA	2.8V@ 0.4mA	NA	NA	NA
AI 17201874	CHPT-88805-18-16	NA	2.8V@ 0.4mA	NA	NA	NA
2M-微距 17201867	GOI2H3-C24TA	0X20_0X21	2.8V@ 40mA	1.8V@ 70mA	NA	NA
2M-微距 17201868	GOI2H3-C24TA	0X20_0X21	2.8V@ 40mA	1.8V@ 70mA	NA	NA

激光对焦

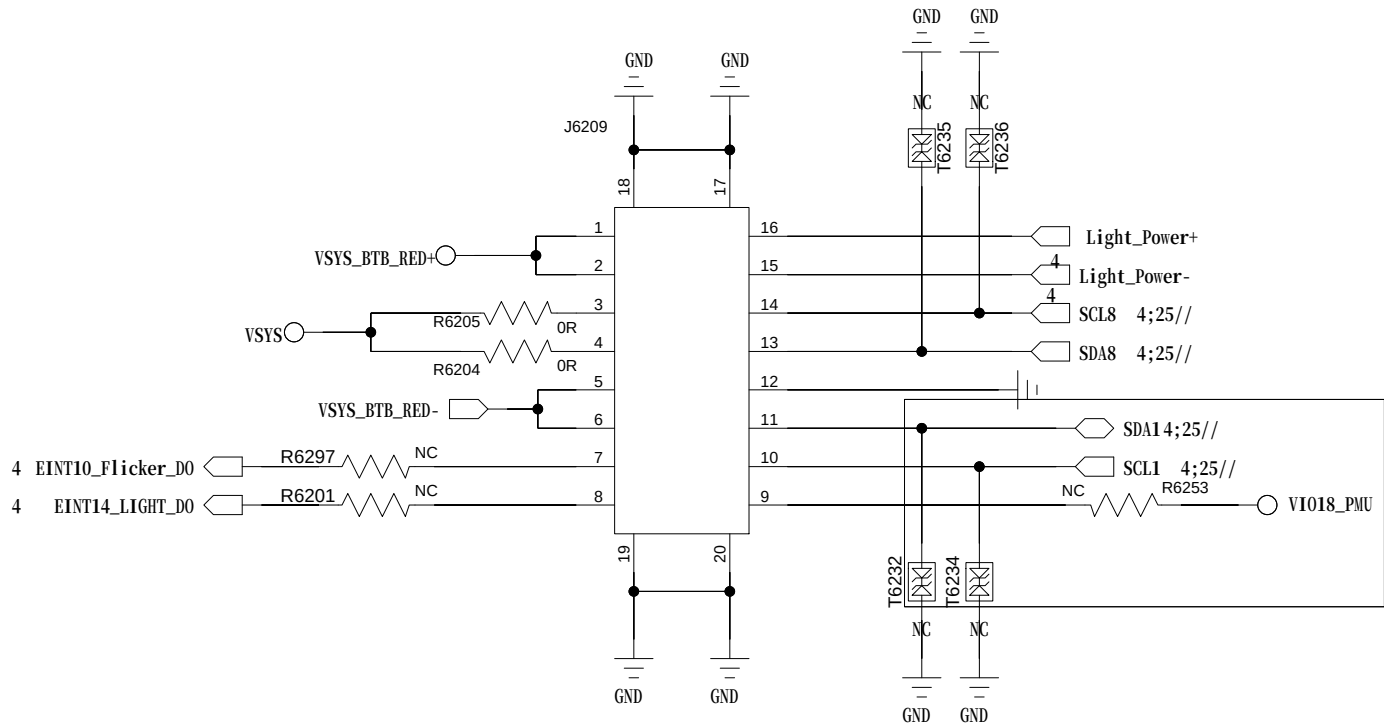
兼容连接器设计

功能1：红外

功能2：FLICKER SENSOR+背出光感

功能3：呼吸灯效

FLICKER SENSOR跟背出光感共用I2C1



TOF

17201739

VI5300

sensor:0xD8(W)/0xD9(R)

3.3V;18mA

- 1、pin1、2、4、5是红外功能
- 2、pin7、8、9、10、11、17、18是背光感、Flicker sensor的功能
- 3、pin13、14、15、16是背光感、Flicker sensor的功能

COMPANY: TRANSSION HOLDINGS

MODEL:G99_STD_MAIN

Modified Date:

31/03/2025:17:52

DRAWN

DATED

TITLE: 62_PERI_CAMERA_II

VERSION:

SHEET: 23 OF 29

CHECKED

DATED

< >

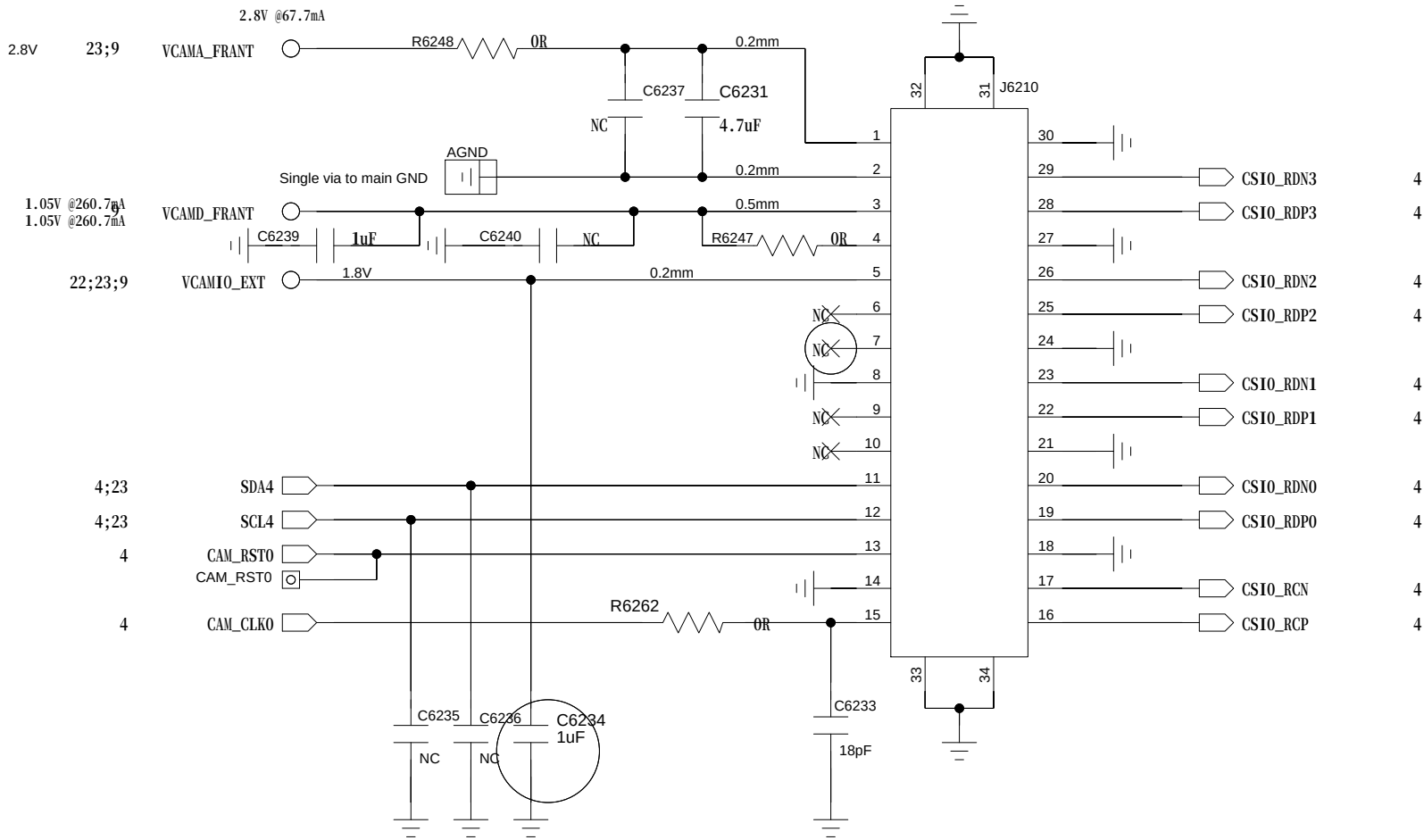
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PERI_CAMERA_III

FRONT MAIN CAMERA (32MFF\13MFF\8MFF)

REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:



SENSOR	VCAMA	VCAMD	VCAMIO	VCAMAF(PIN9)	I2C Address	SAP	PIN4
32M FF GC32E1-WA1XA P24C64F-AAH-1HR	2.8V@55mA	1.2V@ 120mA	1.8V@5mA	NC	Sensor Write:0x94 Read:0x95 EEPROM Write:0xA2 Read:0xA3	17201856	0R
32M FF OV32D40-2A P24C64F-AAH-1HR	2.8V@53mA	1.2V@117mA	1.8V@1mA	NC	Sensor Write:0x6C Read:0x6D EEPROM Write:0xA0 Read:0xA1	TB0	0R
13M GC13A0-WA1XA GT24P64F-XSML1-1R	2.8V@ 45mA	1.2V @115mA	1.8V @ 5mA	NC	Sensor Write:0x72 Read:0x73 EEPROM Write:0xA2 Read:0xA3	17201930	0R
8M G008A8-WADD0	2.8V@40mA	1.2V@110mA	1.8V@2mA	NC	Sensor Write:0x24 Read:0x25	17201890	NC

COMPANY: TRANSSION HOLDINGS				MODEL:G99_STD_MAIN		Modified Date: 03/03/2025:14:06	
DRAWN		DATED		TITLE: 62_PERI_CAMERA_III		VERSION:	SHEET: 24 OF 29
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PERI_SENSORS

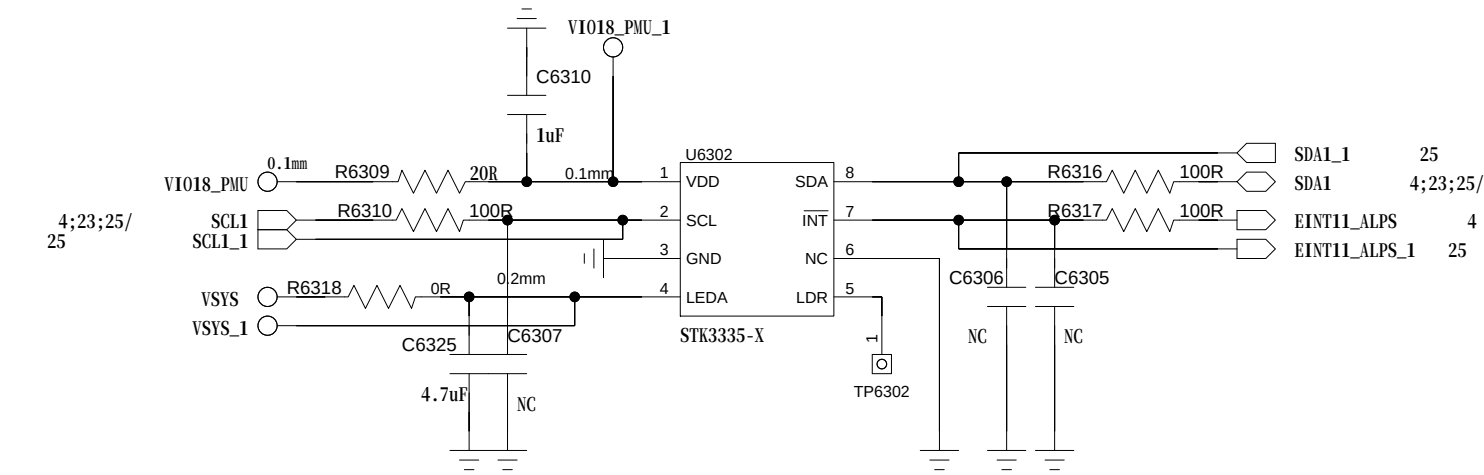
REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:

AL& PS Sensor

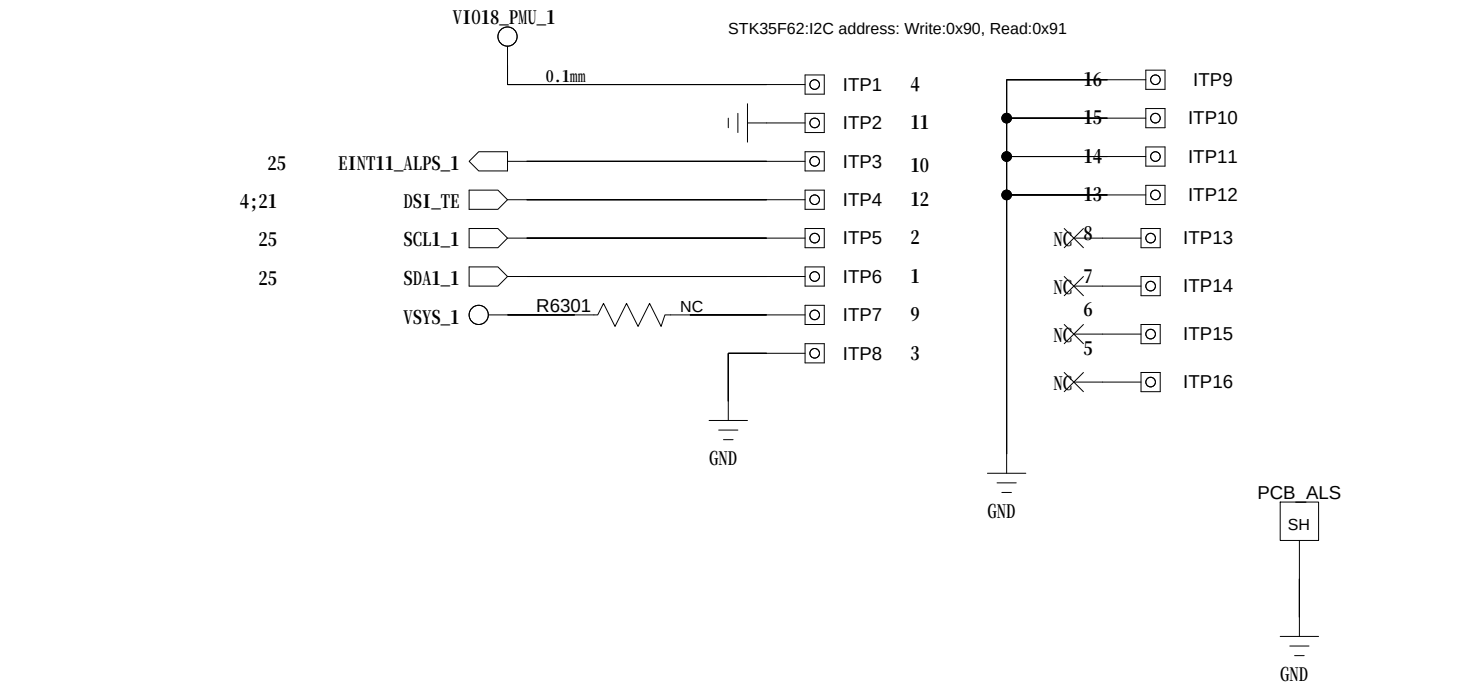
STK33562:I2C address: Write:0x8C, Read:0x8D MN79911:I2C address 0X40: Write:0x82, Read:0x83
LTR-569ALS-WA:I2C address: Write:0x46, Read:0x47
MN78911D:I2C address: Write:0x82, Read:0x83

STK33562/STK3335-X:I2C address 0X46: Write:0x8C, Read:0x8D
注意OLED屏幕光距感垫高板，屏下单光感的时候VSYS串联电阻需要空贴
如果是屏下光距感，一定要看下垫高板原理图和PCB的pin网络是正确匹配
现有的垫高板是屏下单光感，垫高小板的ITP7引脚是接地的，跟主板不一样

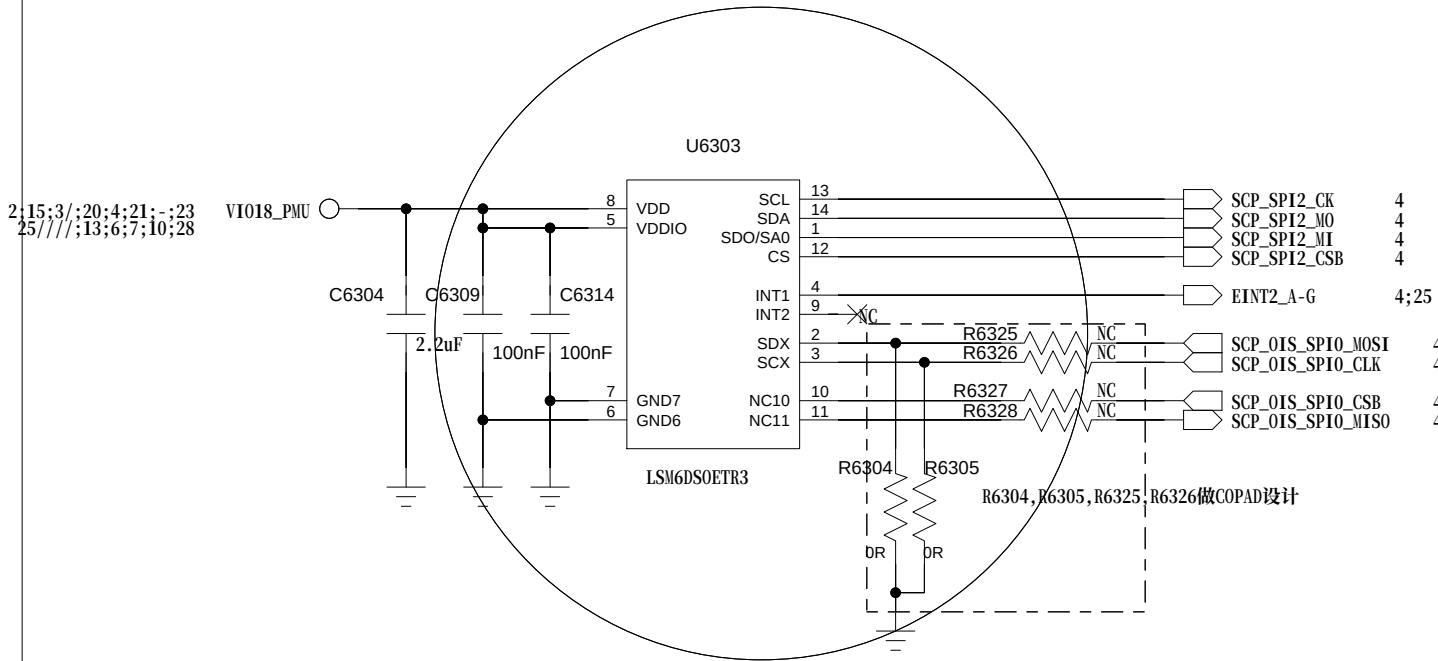
R6309电阻不建议换为0R，主要为了降低电源带来的底噪问题



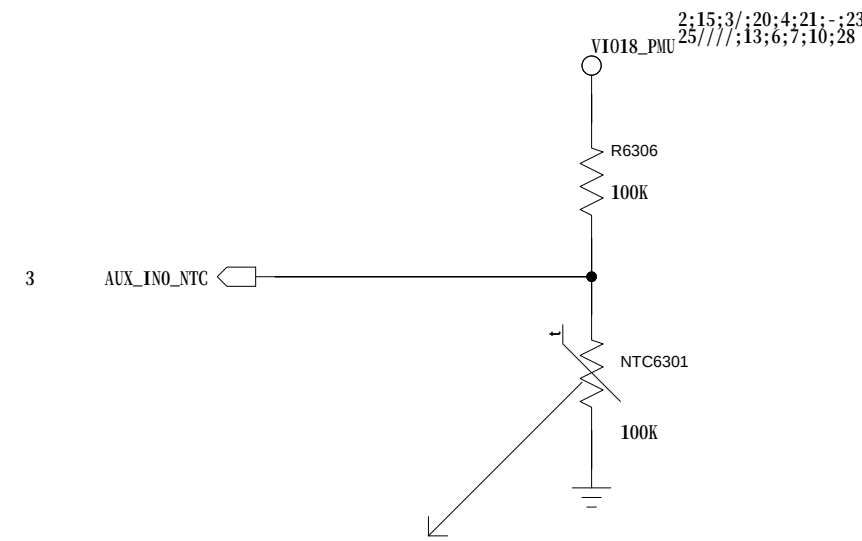
STK3335-X default
MN79911DKDN
MN78912DKDN
STK33562
LS9801



G-Sensor + Gyro Sensor



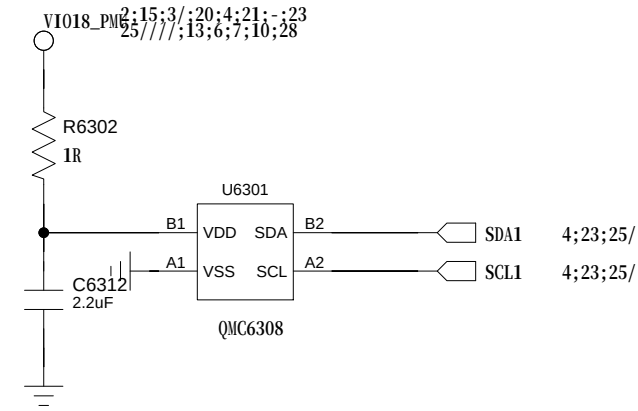
Thermistor to sense AP temperature



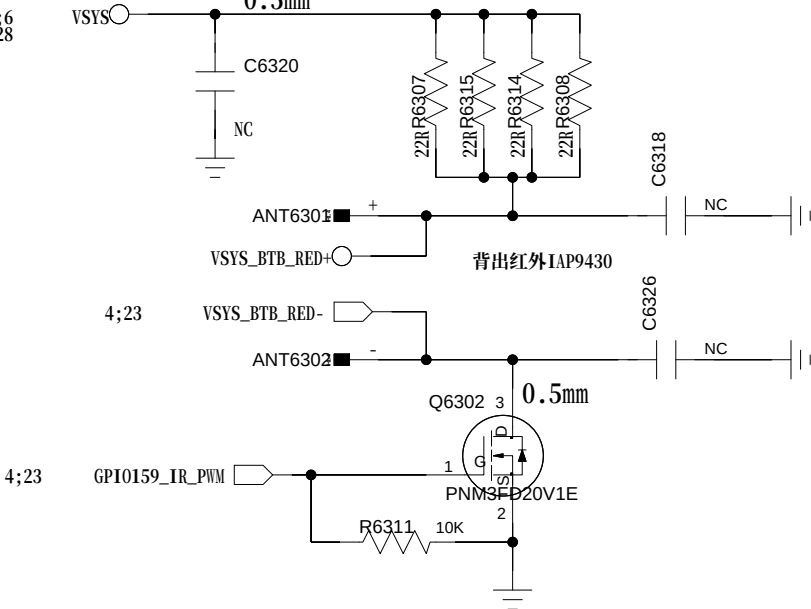
- 1. NTC6301must keep a distance about 6-8 mm away from BB and far from other heat sources 10 mm at least.
- 2. The distance is the shortest distance from package edge to edge.

M-Sensor(COMPASS)

MMC5603:I2C ADDRESS:0x60(Write)/0x61(Read)
QMC6308:I2C ADDRESS:0x58(Write)/0x59(Read)

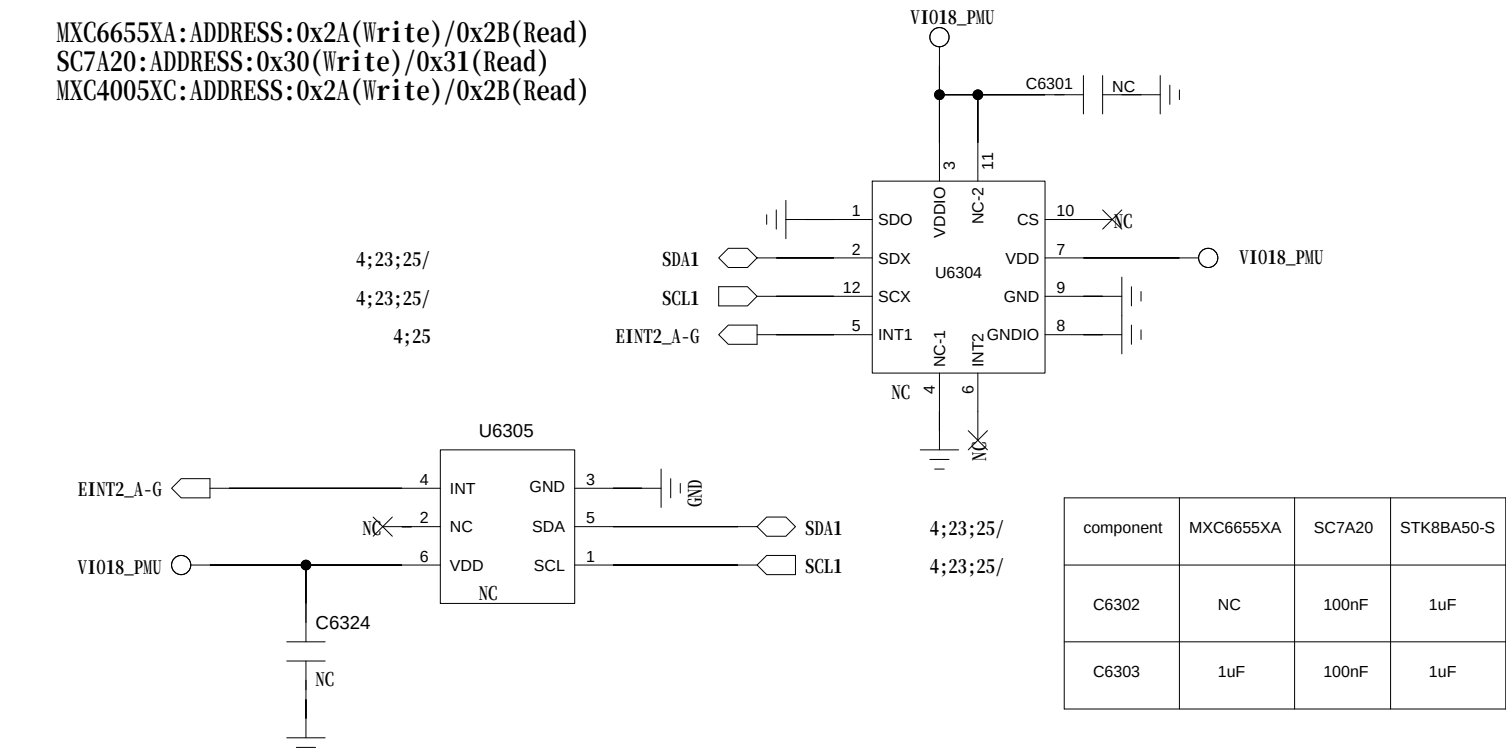


红外



G-Sensor

MXC6655XA:ADDRESS:0x2A(Write)/0x2B(Read)
SC7A20:ADDRESS:0x30(Write)/0x31(Read)
MXC4005XC:ADDRESS:0x2A(Write)/0x2B(Read)



component	MXC6655XA	SC7A20	STK8BA50-S
C6302	NC	100nF	1uF
C6303	1uF	100nF	1uF

COMPANY: TRANSSION HOLDINGS

MODEL:G99_STD_MAIN

Modified Date: 05/03/2025:15:23

DRAWN

DATED

TITLE: 63_PERI_SENSORS

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PERI_EXCH_I0

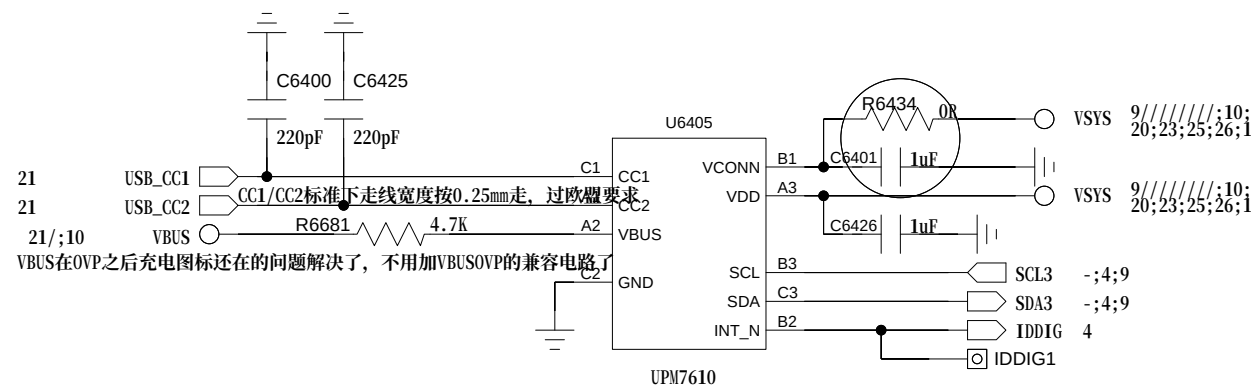
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RT1715:I2C ADDRESS:0x9C(Write)/0x9D(Read)

UPM7610:I2C ADDRESS:0x9C(Write)/0x9D(Read)



GPIO43_WET_X	GPIO44_WET_Y	
H	H	SUB10GND@μç
	L	0y3E
L	H	0y3E
	L	SUB10VBUS@μç

COMPANY: TRANSSION HOLDINGS				MODEL:G99_STD_MAIN		Modified Date: 03/03/2025:14:06	
DRAWN		DATED		TITLE: 64_PERI_EXCH_I0		VERSION:	SHEET: 26 OF 29
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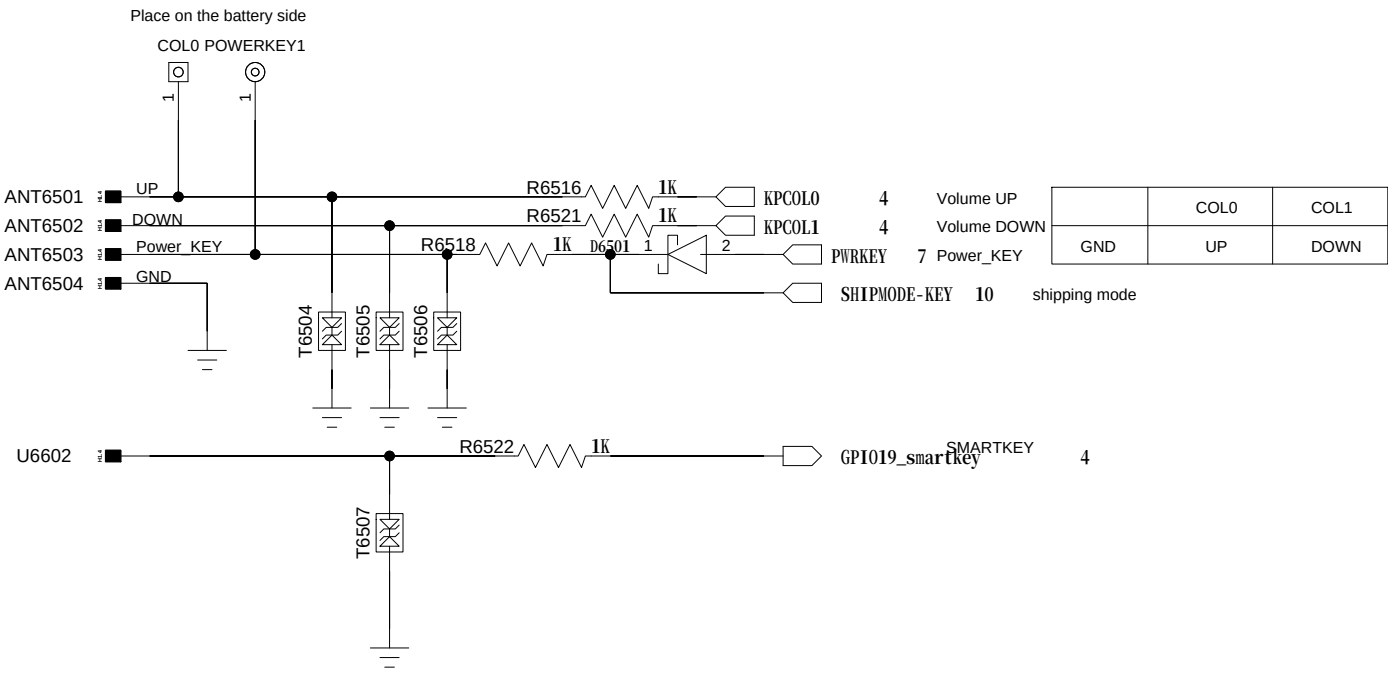
PERI_SIM_SD_KEYPAD

REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:

SIM1/2-SD-CARD

灯效IC电路

SIDEKEY



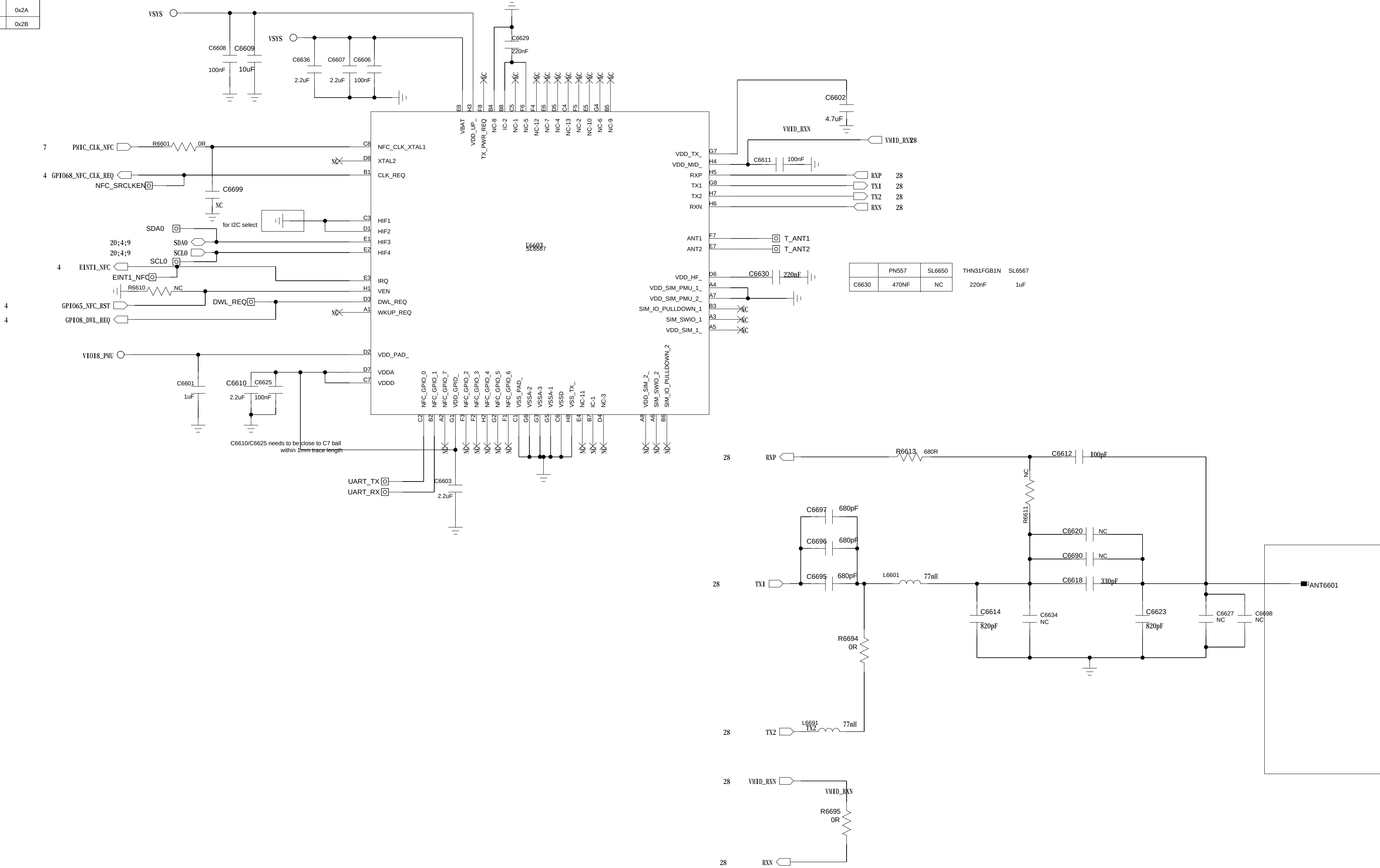
COMPANY: TRANSSION HOLDINGS				MODEL:G99_STD_MAIN		Modified Date: 03/03/2025:14:06	
DRAWN		DATED		TITLE: 65_SD_SIM_KEY		VERSION:	SHEET: 27 OF 29
CHECKED		DATED	< >	Confidentiality	CONFIDENTIAL		

PERI_NFC

SL6550/SL6567***I²C** address: Write:0x50, Read:0x51

HIF2	HIF1	Write	Read	Address
0	0	0x50	0x51	0x28
0	1	0x52	0x53	0x29
1	0	0x54	0x55	0x2A
1	1	0x56	0x57	0x2B

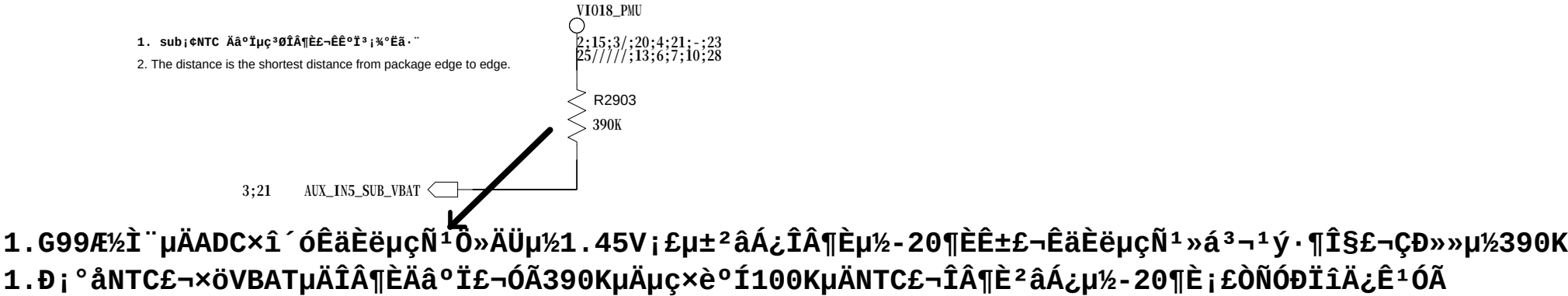
REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:



COMPANY: TRANSSION HOLDINGS				MODEL:G99_STD_MAIN		Modified Date: 03/03/2025:14:06	
DRAWN		DATED		TITLE: 66_NFC		VERSION:	SHEET: 28 OF 29
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REVISION RECORD			
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Thermistor to sense U2704 temperature



COMPANY: TRANSSION HOLDINGS				MODEL:G99_STD_MAIN		Modified Date: 03/03/2025:14:06	
DRAWN		DATED		TITLE: 29_NTC		VERSION:	SHEET: 029
CHECKED		DATED	< >	Confidentiality	CONFIDENTIAL		

